Competing Generalized Cross-Validation Minima and Sensitivity of Adjusted Treatment-Effect Estimates

A Full-Field Penalized Spline Study of On-Farm Strip Trials

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Manuscript working draft — November 2026; Stage 4A refinement incorporated

Abstract

Penalized spline regression is widely used to adjust treatment contrasts for smooth spatial variation, but the smoothing parameter selected for a nuisance trend can influence an unpenalized scientific coefficient. We examine this interaction in an on-farm strip trial consisting of 4,776 observations at 199 Easting positions, with 24 harvest passes at each position. The pointwise model combines a binary product indicator with a penalized cubic B-spline in Northing. We first construct the spline and generalized cross-validation (GCV) criterion independently and evaluate the resulting score landscapes across the field. We then study automatic GCV selection under a common generalized additive model specification, comparing each automatic fit with a 600-point fixed-parameter grid and the exact linear-smooth limit. Under the prespecified numerical detection rule, 28 Eastings exhibit multiple interior GCV minima. At 16 Eastings, the automatically selected fit has a higher GCV score than an evaluated fixed-parameter alternative. We subsequently applied bracketed one-dimensional optimization to all 218 grid-detected interior minima. All local refinement checks passed, and the 16 inferior automatic selections and 199 interior-versus-linear regime classifications were unchanged. The implications for the unpenalized product coefficient are heterogeneous: a substantial score difference may correspond to a negligible coefficient change, whereas a smaller score difference may accompany an appreciable change. Changes in the criterion-preferred smoothing regime can also occur without a detectable optimization discrepancy. These results provide a reproducible computational case study distinguishing criterion geometry, optimization performance, and coefficient sensitivity. They do not establish global GCV optima, identify the true spatial treatment-effect function, or demonstrate that spatial dependence causes the observed behavior.

Keywords: generalized cross-validation; penalized cubic B-splines; generalized additive models; smoothing parameter selection; partially linear models; treatment effects; spatial field trials; numerical optimization.

Manuscript scope: This first paper covers the completed data audit and computational experiments through Stage 4A. Stage 4A refines previously detected grid candidates; it is not a complete stationary-point enumeration. The proposed Reiss–Ogden h-function analysis is excluded. Any analysis of first-stage preprocessing sensitivity is outside the present numerical results and remains a designated revision placeholder.

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1 Introduction

1.1 Motivation: a scientific coefficient embedded in a smoothing problem

Large on-farm strip trials generate dense spatial measurements that offer opportunities to examine how treatment responses vary over agricultural fields. They also expose an important statistical difficulty: a difference between product treatments must be distinguished from variation in soil, management, topography, and other spatial processes. Smoothing methods are natural tools for this purpose because the environmental component of a yield surface need not follow a simple linear function. Long-established work on agricultural field experiments explicitly studies estimation of treatment effects jointly with smoothing of spatial trends (Green, 1985; Green et al., 1985), and more recent analyses use penalized splines or locally weighted models to study spatial heterogeneity (Rodríguez-Álvarez et al., 2018; Rakshit et al., 2020).

Our motivating analysis estimates a treatment contrast independently at successive Easting locations. At each Easting, a binary product indicator enters the model parametrically, while the Northing trend is represented by a penalized cubic spline. It is tempting to regard these as separate operations: a flexible smoother accounts for background spatial structure, leaving the product coefficient to measure the adjusted contrast. Algebraically, however, they are coupled. Changing the smoothing parameter changes the fitted nuisance component and can change the coefficient of the unpenalized treatment indicator. This coupling is familiar in partially linear and semiparametric regression, where smoothing bias can propagate into an accompanying finite-dimensional parameter (Rice, 1986; Speckman, 1988).

The practical question is therefore not solely how closely the smoother fits the yield profile. We also need to know whether the numerical smoothing-parameter selection procedure is finding a comparatively favorable value of its criterion, how the criterion behaves across the parameter domain, and whether choosing different smoothing regimes has consequences for the scientific coefficient. These questions have different answers and should not be collapsed into one diagnosis of "oversmoothing" or "undersmoothing."

1.2 Pointwise regression formulation

Let e index an observed Easting and let $i = 1, \ldots, 24$ index the harvest passes at that Easting. Write $Y_i(e)$ for the recorded yield, $P_i = \mathbf{1}\{\text{Product}_i = B\}$ for the treatment indicator (Product E is the reference group), and N_i for the pass Northing. The working pointwise model is

$$Y_i(e) = \beta_0(e) + \beta_B(e)P_i + f_e(N_i) + \varepsilon_i(e). \tag{1}$$

Here $\beta_B(e)$ is the Northing-adjusted B-minus-E treatment coefficient. The function f_e accounts for Northing-dependent yield variation at the same Easting. A second-derivative roughness penalty discourages unnecessary curvature in f_e but does not directly penalize $\beta_B(e)$.

For a fixed spline design $\mathbf{B}$, a representative penalized least-squares formulation is

$$\min_{\beta_0,\beta_B,\boldsymbol{\theta}} \left\{ \sum_{i=1}^{24} \left[Y_i(e) - \beta_0 - \beta_B P_i - (\mathbf{B}\boldsymbol{\theta})_i \right]^2 + \lambda_e \boldsymbol{\theta}^\top \mathbf{S} \boldsymbol{\theta} \right\}, \qquad \lambda_e > 0. \tag{2}$$

The penalty $\mathbf{S}$ represents curvature; λ_e sets its weight. As λ_e increases, the penalized nonlinear component is progressively suppressed, and the fitted Northing trend approaches the penalty's linear null space. In our software experiments the tuning argument is called `sp`; with one penalized smooth, `sp` is the smoothing parameter λ_{mgcv}. Its numerical value should not be equated with the smoothing parameter of an independently constructed spline basis and penalty.

Generalized cross-validation (GCV) combines residual error with an effective-degrees-of-freedom adjustment. For our Gaussian pointwise fits and the score normalization used in the completed computations, it is

$$\mathrm{GCV}_e(\lambda) = \frac{n\ \mathrm{RSS}_e(\lambda)}{\{n - \mathrm{EDF}_e(\lambda)\}^2}, \qquad n = 24, \tag{3}$$

where $\mathrm{EDF}_e(\lambda)$ includes the parametric terms and the smooth. A lower GCV score is preferred *according to the criterion*. That does not make the corresponding treatment effect an unbiased estimate of an unknown truth.

1.3 Why the criterion landscape matters

Multiple local extrema in smoothing-selection criteria are already documented, including explicit penalized-spline examples and algorithms for searching them (Li and Cao, 2023). Modern GAM software documentation also explicitly warns that GCV scores can have local minima separated by flatter regions, and that numerical minimization may stop at a local solution (Wood, 2026). Reiss and Ogden (2009) analyze stationary-point equations for GCV and restricted maximum likelihood (REML) in a defined class of semiparametric linear models, showing why more than one candidate solution must be considered. More recent work has developed systematic smoothing-parameter search intervals to help avoid undesirable local optima (Li and Cao, 2023).

These studies establish that multiple minima and local optimization difficulties are *known* features of smoothing-parameter selection. We do not propose them as new discoveries. The narrower question of the present study is how such features appear across a regularly indexed series of related harvest-strip fits, and how strongly alternative choices affect an unpenalized product coefficient. Repeated pointwise fits are especially instructive because the design is held fixed while the response changes spatially.

For clarity, we distinguish three numerical comparisons. First, the *criterion landscape* $\mathrm{GCV}_e(\lambda)$ describes available smoothing regimes, including multiple local minima. Second, the *optimizer gap* compares the automatically selected score with the lowest score found among evaluated grid points:

$$D_{\text{grid}}(e) = \mathrm{GCV}_{e,\text{auto}} - \min_{\lambda \in \mathcal{G}} \mathrm{GCV}_e(\lambda). \tag{4}$$

Third, the *coefficient discrepancy* compares the associated treatment estimates:

$$\Delta_{\beta,\text{grid}}(e) = \widehat{\beta}_{B,\text{auto}}(e) - \widehat{\beta}_{B,\text{best grid}}(e). \tag{5}$$

The criterion landscape, its evaluated score contrast, and its coefficient contrast are logically distinct. A small automatic-versus-*grid* coefficient difference can reflect discretization of a continuous criterion rather than an inferior automatic optimum. Stage 4A therefore adds refined and exact-linear reference candidates; the reference sets are defined explicitly in Table 2.

1.4 Dataset and completed computational study

The complete harvest dataset contains 4,776 observations distributed over 199 Easting positions, from 2 to 398 in increments of 2, and 24 passes at each position. Product assignments and Northing coordinates are fixed by pass across Eastings. The initial full-field data audit reconciled the original three diagnostic profiles at Eastings 214, 216, and 218 with the complete dataset. This common experimental design allows a manual spline basis and penalty to be shared across response vectors, whereas the yield profiles themselves vary with Easting.

The completed study comprises two separate spline implementations. First, a manual Base R reconstruction evaluates a cubic B-spline penalized least-squares system over 400 log-smoothing-parameter values at each Easting, providing 79,600 GCV evaluations. Second, the original GAM specification is fitted with `mgcv` using a penalized cubic B-spline of basis dimension 10 and a Gaussian response. Automatic GCV selection is compared with a 600-point fixed log-smoothing-parameter grid at each Easting, providing 119,400 fixed-parameter GAM fits. The manual and package-based calculations address closely related model structures, but their finite-parameter smoothing scales are not directly interchangeable because their basis and penalty constructions are not identical.

The second implementation yields 28 Eastings with at least two grid-detected interior local minima. At 16 Eastings the automatic GCV score exceeds the score of an evaluated grid candidate by the prespecified numerical tolerance. Among these cases, the treatment-coefficient discrepancy varies considerably. The contrast between Eastings 174 and 374 is illustrative: the former shows a relatively modest GCV score gap coupled with a larger change in the Product B coefficient, while the latter shows a larger criterion gap but a comparatively small coefficient difference. Other locations exhibit a change in the criterion-preferred smoothing regime without a detectable optimizer deficit. These are numerical observations *within one field and one set of evaluated models*; they are not evidence that the alternative coefficients are closer to the unknown true effects.

1.5 Contributions, scope, and organization

The paper makes three deliberately bounded contributions. First, it documents GCV profiles over an entire spatially indexed set of pointwise treatment-adjusted fits, rather than presenting only a few selected curves. Second, it separates three phenomena that can look similar in a treatment-effect plot: competing stationary candidates, an automatic solution that is inferior to an evaluated alternative, and a change in the criterion-preferred smoothing regime between neighboring datasets. Third, it quantifies the resulting changes in an unpenalized treatment coefficient, showing that the magnitude

of a GCV score difference does not, by itself, determine the magnitude of a coefficient discrepancy.

The study does not claim a new global optimizer, a complete enumeration of stationary points, or a mathematical characterization of the GCV surface. The observed grid minima are numerical candidates; a finite grid does not prove global optimality. In addition, the pointwise analyses do not establish the underlying spatial error covariance or the provenance of any preprocessing smooth applied to the recorded yields. These issues matter for inferential interpretation and are discussed as limitations rather than treated as resolved explanations.

Section 2 reviews spline regression, smoothing-parameter selection, partially linear models, and spatial field-trial methodology. The subsequent sections describe the harvest data, model implementation, full-field computational results, and Stage 4A numerical refinement. An analytic stationary-point enumeration and additional simulation studies are outside the completed empirical work.

2 Literature Review

2.1 From spline approximation to penalized additive regression

A spline is a piecewise polynomial function whose smoothness at its knots is controlled by the polynomial order and knot multiplicities. The B-spline representation developed in classical approximation theory is attractive for computation because of local support and stable recursive evaluation (Boor, 2001). In regression settings, a smooth response component is represented as a linear combination $f(x) = \sum_j \theta_j B_j(x)$, so function estimation becomes estimation of a coefficient vector. The mathematical spline space and the numerical basis used to represent that space must be distinguished. Equivalent function spaces can be expressed in different bases, and penalty scaling may change even when the fitted-function family appears similar.

Smoothing splines replace interpolation or unpenalized least squares with a compromise between fit and a roughness functional. The smoothing problem in Craven and Wahba (1979) combines fidelity to data with an integrated derivative penalty, and Wahba (1990) develops the broader theory connecting spline smoothing and regularization. Penalized B-spline approaches instead work with a chosen finite spline basis and an explicit penalty matrix. Eilers and Marx (1996) popularized the P-spline approach, usually with finite differences of adjacent basis coefficients. Such difference penalties are not numerically identical to integrated derivative penalties in arbitrary bases or for arbitrary knot spacing; these distinctions matter when comparing smoothing parameters across implementations (Boor, 2001; Wood, 2017).

Generalized additive models (GAMs) place smooth components alongside ordinary regression coefficients (Hastie and Tibshirani, 1990; Wood, 2017). Their computation relies on constrained bases, quadratic penalties, and numerical criteria for selecting smoothing parameters. In the current application, the GAM has a single smooth of Northing and an ordinary binary product coefficient. The actual package model is a penalized B-spline term, not the default thin-plate smoother used in many unrelated GAM applications. This specification matters: claims about one basis, penalty, or smoothing parameter cannot automatically be transferred to another.

The linear null space of a second-derivative penalty plays a central role. As the penalty weight grows large, curvature is suppressed while constant and linear components remain unpenalized. For the fitted treatment-plus-Northing model, the corresponding limiting fit is ordinary linear regression with an intercept, a product indicator, and a linear Northing term. This supplies a scientifically interpretable reference against which flexible alternatives can be compared. The limit is shared by the two implemented spline constructions even though their finite-λ parameterizations differ.

2.2 Generalized cross-validation and its optimization

Craven and Wahba (1979) introduced GCV as a data-driven means of selecting a smoothing level in linear smoothing problems, while Golub et al. (1979) placed the same idea in a broader regularization context. The familiar linear-smoother form is a normalized squared residual divided by the squared effective residual degrees of freedom. For the Gaussian fits studied here it reduces to equation (3). The numerator generally grows as penalization increases because the fitted curve loses flexibility, whereas the effective-degrees-of-freedom adjustment rewards parsimony. Consequently, the total GCV score need not vary monotonically with $\log \lambda$ and can have more than one interior local minimum.

The numerical behavior of data-driven smoothing choices has a substantial history. Härdle et al. (1988) study how automatically selected smoothing parameters compare with optimal ones, while Li and Cao (2023) directly illustrate distinct local GCV minima and develop search-interval methods. Reiss and Ogden (2009) provide an explicit analytical treatment of GCV and REML stationary-point equations for a particular class of semiparametric linear models. Their theory helps explain the possibility of competing solutions; it is not a theorem that every GAM score has multiple minima, nor does it prove that the lowest value on any particular numerical grid is the global optimum.

GAM-fitting algorithms have become more stable through increasingly direct and efficient smoothing-parameter estimation (Wood, 2004; Wood, 2011). Nevertheless, software convergence is a numerical statement about a fitting algorithm, not a guarantee that every part of a nonconvex criterion landscape has been explored. The `mgcv` documentation itself notes that flat score regions and intervening shallow barriers can yield undesirable GCV local optima (Wood, 2017). An independent fixed-parameter sweep therefore provides a direct and transparent diagnostic: when the automatic score exceeds that of a tested fixed parameter, its selection is inferior *with respect to the same GCV criterion*. No assumptions about the true mean function are needed for this numerical comparison.

A critical contribution of Li and Cao (2023) is the development of automatically determined smoothing-parameter search intervals for penalized splines. Their work already addresses the risk of selecting an undesirable local optimum and the importance of searching an adequate parameter range. For that reason, a claim that grid search itself is a new solution would be untenable. Our completed investigation instead uses a fixed grid as a diagnostic instrument to compare criterion scores at all Eastings and to identify whether those differences have noticeable consequences for the unpenalized product contrast. This also imposes a limitation: without a separately established search interval or exhaustive stationary-point analysis, the lowest observed grid score is not certified as a global minimum.

2.3 GCV versus likelihood-based smoothing selection

GCV is not the only available smoothness-selection criterion. Marginal likelihood and restricted maximum likelihood (REML) have long been considered alternatives, and their behavior can differ in finite samples and under model misspecification. Wahba (1985) compares GCV with generalized maximum likelihood for smoothing splines. Reiss and Ogden (2009) analyze mathematical relationships between the selection equations for GCV and REML, while Wood (2011) develops stable direct REML/ML smoothing selection for generalized semiparametric models. This literature supplies plausible alternatives to GCV but does not justify a blanket claim that REML is preferable in every setting.

In particular, Berry and Helwig (2021) compare cross-validation, information-criterion, and likelihood-based smoothing parameter choices across simulated conditions and emphasize that performance depends on the data-generating mechanism. Our completed experiments were designed to characterize the GCV criterion and compare automatic GCV with alternative values of *that same criterion*. REML, ML, or alternative cross-validation selectors were not evaluated as new full-field experiments. Their merits therefore belong to the context and future-work discussion, not to the empirical claims of this paper.

2.4 Partially linear models and the unpenalized coefficient

The model in equation (1) is a pointwise partially linear model: an ordinary regression coefficient coexists with a nonparametric function. This is an established semiparametric setting. Rice (1986) shows that the parametric part of a partially splined model can have non-negligible smoothing bias. Speckman (1988) studies bias and variance for parametric estimates in partially linear models and explains why partial-residual methods can improve bias properties under suitable assumptions. These results are important because a smoothing parameter chosen to favor prediction or recovery of a nuisance trend need not be optimal for estimation of a specific scientific coefficient.

The dependence of an unpenalized coefficient on smoothing can be seen directly from the penalized normal equations. Let $\mathbf{p}$ denote the product indicator, and let $\mathbf{W}$ contain an intercept and the spline columns, with penalty $\mathbf{S}_W$ that is zero on unpenalized components. For a fixed λ, define

$$\mathbf{M}_\lambda = \mathbf{I} - \mathbf{W}(\mathbf{W}^\top\mathbf{W} + \lambda\mathbf{S}_W)^{-1}\mathbf{W}^\top. \tag{6}$$

Whenever the indicated inverses exist, eliminating the nuisance coefficients yields

$$\widehat{\beta}_B(\lambda) = \left(\mathbf{p}^\top\mathbf{M}_\lambda\mathbf{p}\right)^{-1}\mathbf{p}^\top\mathbf{M}_\lambda\mathbf{y}. \tag{7}$$

These are standard partitioned penalized least-squares identities, not a new theorem. They make the relevant mechanism explicit: penalizing the nuisance smooth changes $\mathbf{M}_\lambda$, which changes the adjusted treatment coefficient unless special orthogonality conditions eliminate that dependence. Our empirical coefficient comparisons are illustrations of this dependence, not estimates of bias relative to a known underlying effect.

The pass-level treatment arrangement also permits an observable design check. Residualizing the

12 B indicators against an intercept alone gives $\mathbf{p}^\top \mathbf{M}_{\text{int}} \mathbf{p} = 6$; residualizing against the intercept and *linear* Northing gives 5.874886, or 97.9% of the intercept-only quantity. This shows modest overlap with the *linear* spatial trend but does not assess nonlinear overlap. For the independently constructed Stage 2 cubic B-spline and its original 400-point penalty grid, direct evaluation of the expression above gives $\mathbf{p}^\top \mathbf{M}_\lambda \mathbf{p}$ ranging from 5.117861 at $\log \lambda = -1$ to 5.874886 at $\log \lambda = 23$; the design is fixed over all 199 Eastings. A separate Python implementation reproduced the saved Stage 2 Easting-214 GCV scores to within 4.7×10^{-9} and its product coefficients to within 1.5×10^{-9} (see the archived diagnostic code and CSV). These are *manual-basis design quantities*, not reported values for the distinct `mgcv` spline, and they are not standard errors or a formal information calculation under unspecified spatial covariance.

This distinction also prevents a common interpretive error. An automatically chosen smoothing parameter can have an inferior GCV score but produce essentially the same estimated treatment coefficient as a lower-scoring alternative; conversely, close GCV scores can correspond to distinct adjusted coefficients. GCV evaluates a whole fitted model under a prediction-oriented tradeoff. It does not directly minimize the absolute difference between the estimated product coefficient and an unknown treatment effect. The size of a criterion gap, the sensitivity of the coefficient, and statistical truth are therefore separate concepts.

2.5 Spatial field-trial analysis and treatment-effect estimation

Agricultural field experiments have a long history of treatment-effect estimation with spatial adjustment. Green (1985) formulates field-trial analyses as models in which spatial smoothing and treatment estimation are considered together. This establishes that our overall modeling strategy is not new. More recently, Rodríguez-Álvarez et al. (2018) develops P-spline adjustment for spatial heterogeneity in plant-breeding experiments, illustrating the continuing importance of flexible spatial trends in agricultural analysis.

For on-farm trials, Rakshit et al. (2020) develops methods for spatially varying treatment effects in a geographically weighted regression framework. This is especially close to our application domain: local estimation, a treatment factor, and selection of a spatial smoothing or bandwidth scale all enter the analysis. The distinction between geographically weighted regression and the separate pointwise penalized GAMs examined here is relevant, but it is not by itself a guarantee of methodological novelty. Claussen (2025) analyzes on-farm strip trials through functional-data and spatial approaches; the current study follows a related strip-trial setting while isolating numerical smoothing-selection behavior.

Our completed work is intentionally narrower than a comprehensive spatial treatment-effect analysis. We fit separate models at a series of Eastings, holding the pass-level covariate arrangement fixed. We then inspect the GCV criterion and compare estimates of the product contrast across distinct smoothing choices. A change in the coefficient trajectory between neighboring Eastings may reflect genuine spatial treatment variation, changing spatial nuisance structure, smoothing-regime selection, or some combination of these. Without a known ground truth or a specified joint spatial covariance model, the present experiments cannot assign those changes uniquely to a single cause.

2.6 Spatial dependence as a limitation, not an established mechanism

The GCV criterion is conventionally motivated under error assumptions that can be inappropriate for spatially dependent observations. J. Opsomer et al. (2001) reviews the sensitivity of smoothing-parameter selection to correlated errors; Krivobokova and Kauermann (2007) studies this problem in penalized spline regression and compares selection approaches under correlated data; and Francisco-Fernández and J. D. Opsomer (2005) develops a correlation-adjusted GCV procedure for spatially correlated nonparametric regression. This literature provides a credible reason to examine spatial dependence in our trial, especially given repeated passes and potentially smoothed yield inputs.

But an important epistemic boundary remains. The presence of multiple GCV minima or a higher-scoring automatic solution does not establish that correlated errors caused either feature. Likewise, an apparent change in the spatial product-coefficient trajectory cannot be labeled a spurious artifact without an appropriate comparison model and uncertainty analysis. Our full-field grid calculations directly reveal *criterion geometry and numerical comparisons*; they do not estimate a spatial covariance function or test a dependence-correction procedure. We therefore treat the correlated-error literature as relevant context and an explicit limitation, rather than using it as an unsupported causal explanation.

2.7 Position of the present study relative to existing work

Several aspects of our problem have already been investigated independently: GCV as a smoothness selector (Craven and Wahba, 1979), multiple smoothing-criterion optima (Reiss and Ogden, 2009), numerical search intervals (Li and Cao, 2023), effects of smoothing on semiparametric coefficients (Rice, 1986; Speckman, 1988), and spatial adjustment of agricultural treatment effects (Green, 1985; Rakshit et al., 2020). The contribution of this paper is not priority in any one of those topics.

Instead, its focus is a documented, reproducible intersection of the topics. A common pass-level experimental design allows a series of pointwise penalized GAMs to be examined under a fixed model specification. The complete fixed-parameter grid makes the local GCV landscapes visible. Comparisons against the automatic solution distinguish optimization discrepancies from criterion-preference changes. Finally, simultaneous tracking of the unpenalized product coefficient identifies which numerical discrepancies matter for the reported scientific contrast. This supplies a systematic empirical case study and a motivated set of questions for subsequent theory and simulation, without claiming that a case study alone settles them.

3 Harvest Trial and Data Structure

3.1 Spatial indexing and observational units

The supplied harvest-trial results are organized by Easting and harvest pass. Let $\mathcal{E} = \{2, 4, \ldots, 398\}$ denote the 199 observed Easting coordinates and $i \in \{1, \ldots, 24\}$ identify a pass. The recorded response can thus be indexed as $Y_i(e)$ for $e \in \mathcal{E}$. The Stage 1A audit reports one observation for every Easting–pass combination, giving $199 \times 24 = 4,776$ records. At each Easting, the 24

Table 1: Observed structure of the on-farm strip-trial dataset. Counts and coordinate ranges are derived from the uploaded Stage 1A audit outputs; pair assignments and coordinate limits are cross-referenced against the Stage 1B matched observations.

Characteristic	Observed value
Yield observations	4,776
Distinct Easting positions	199
Easting coordinate sequence	$2, 4, \ldots, 398$
Passes at each Easting	24
Product E / Product B passes per Easting	12 / 12
Distinct pass-pair identifiers	12
Passes within each pair	2
Observed Northing coordinate range	0.7831 to 140.6811
Easting–pass combinations	4,776 unique
Within-pass Product, Pair, and Northing across Eastings	Constant

records are balanced between Product B and Product E, with 12 records in each group. Table 1 characterizes this data arrangement.

The *Easting* coordinate identifies the location at which each pass's yield is recorded, whereas *Northing* indexes the pass's location orthogonal to the sequence of Easting positions. The Stage 1A pass-design audit confirms that every pass retains one Northing value, one Product label, and one Pair identifier over the observed Eastings. Accordingly, for a given pass i, N_i and P_i do not change with e. This allows the model at each Easting to be viewed as having a common covariate arrangement but a different response vector. Figure 1 displays the observed pass-specific layout reconstructed from the audit and matched reference observations.

3.2 Treatment assignment and paired passes

The 24 pass identifiers form 12 recorded pairs. Each pair comprises two passes assigned the same Product, and the Product assignments alternate between successive pairs in the matched reference observations: two E passes, then two B passes, repeating across the ordered Northing locations. Each Easting thus provides the same sequence of treatment indicators P_i and Northing coordinates N_i, not a fresh allocation of products. The available audit outputs establish this arrangement but do not, by themselves, establish the mechanism by which products were assigned or whether allocation was randomized. We therefore describe the *observed treatment arrangement* without making a randomization-based claim.

The repeated pass geometry has a direct statistical consequence. Since the product indicator is associated with specific Northing positions, estimating an adjusted treatment contrast requires distinguishing a systematic B-versus-E difference from a potentially nonlinear Northing trend. The model in equation (1) makes that adjustment through a smooth nuisance function f_e while retaining the parametric Product coefficient of scientific interest. The same product assignment is present at all Eastings; therefore, changes in the estimated coefficient across Eastings cannot be ascribed to changes in which pass received which product.

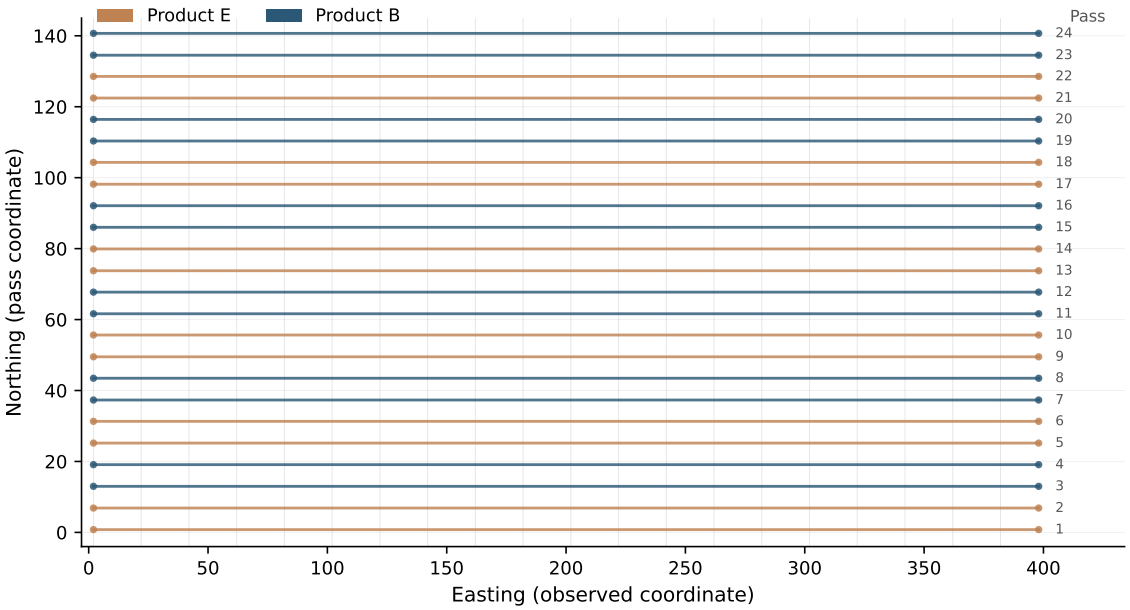


Figure 1: Observed harvest-pass and treatment layout. Each horizontal line is one pass observed at the 199 Easting coordinates from 2 through 398. Product labels and Northing coordinates are held fixed within pass. The displayed layout is reconstructed from `Easting_Audit.csv`, `Pass_Design_Audit.csv`, and the pass-matched records in `Stage1B_Row_Level_Reconciliation.csv`. It visualizes the recorded design, not a randomized-allocation procedure or independently observed replicate fields.

3.3 Correspondence with the original three-Easting analysis

The initial investigation used separately saved data for Eastings 214, 216, and 218. Stage 1B matched those records to the corresponding observations of the full-field dataset by Easting and `PassNo`. All 72 matched records passed the joint checks of Yield, Northing, Product, Pair, and Pass. Across these three subsets, the maximum absolute Yield differences were 3.013×10^{-12}, 4.007×10^{-12}, and zero, respectively. The Northing differences were zero at the reported precision. The reconciliation supports numerical continuity with the original case study and rules out a change in these input records as an explanation for differences between the three-Easting and full-field calculations. This is a provenance check, not a hypothesis test or evidence about GCV's statistical performance.

3.4 Implications and boundaries of the data structure

For any fixed spline parameterization, the vector of pass covariates is common across Eastings. Writing $\boldsymbol{y}_e = (Y_1(e), \ldots, Y_{24}(e))^\top$, the response array has the conceptual organization

$$\boldsymbol{Y} = [\boldsymbol{y}_2, \boldsymbol{y}_4, \ldots, \boldsymbol{y}_{398}] \in \mathbb{R}^{24 \times 199}. \tag{8}$$

Once the Northing basis is specified, the associated unpenalized columns and penalty matrix can be reused across Eastings within that spline construction. This makes variation in the GCV criterion

across Eastings attributable to changes in the observed response vectors, conditional on the fixed design and model specification. It does not make the 199 fitted models statistically independent: neighboring Eastings involve measurements on the same harvest passes, and the covariance of those measurements has not been established by the Stage 1 audits.

A final qualification concerns the status of Yield as supplied for analysis. The filenames indicate processed harvest-pass data, but the uploaded audits and reconciliations do not document the original measurement instrument or all prior processing steps. In particular, they do not establish whether and how the supplied yield series were smoothed along Easting before the present pointwise fits. We therefore refer to the *recorded or supplied Yield* throughout the empirical analysis and do not assert that these observations are independent, unsmoothed raw harvester readings.

4 Statistical Methods and Computational Experiments

4.1 Stage 2: independent manual spline construction

We first investigated the score geometry of a penalized cubic spline with an implementation independent of automatic `mgcv` fitting. At a given Easting, equation (1) was represented by a common pass-level spline basis. With $n = 24$ unique Northing locations, the construction specified cubic degree $d = 3$ and $k = 10$ B-spline basis functions. The knot sequence comprised six equally spaced interior knots, the two observed Northing endpoints, and three extended knots on either side of the observed range, for a total of 14 knots. The cubic basis was computed using the Cox–de Boor recursion (Boor, 2001), producing $\mathbf{B} \in \mathbb{R}^{24\times 10}$.

For a spline $f(N) = \sum_{j=1}^{10} \theta_j B_j(N)$, the integrated squared-second-derivative penalty was

$$J(f) = \int_{N_{\min}}^{N_{\max}} \{f''(u)\}^2 \, du = \boldsymbol{\theta}^\top \mathbf{S}_s \boldsymbol{\theta}, \qquad (\mathbf{S}_s)_{jk} = \int_{N_{\min}}^{N_{\max}} B_j''(u) B_k''(u) \, du. \tag{9}$$

The penalty entries were assembled using Simpson quadrature on each knot span. The product of two second derivatives of cubic B-splines is piecewise quadratic, so three-point Simpson quadrature integrates this product exactly on each span in exact arithmetic. The resulting 10×10 penalty had numerical rank eight, with unpenalized constant and linear spline directions. This is a continuous curvature penalty, not a finite-difference penalty applied to the coefficients.

Because the spline basis already includes a constant function, the intercept and smooth would be non-identifiable without a constraint. A QR-based orthogonal complement $\mathbf{Z} \in \mathbb{R}^{10\times 9}$ to the vector $\mathbf{B}^\top \mathbf{1}$ enforced $\mathbf{1}^\top \mathbf{B}\mathbf{Z} = 0$. Defining

$$\mathbf{B}_c = \mathbf{B}\mathbf{Z} \in \mathbb{R}^{24\times 9}, \qquad \mathbf{S}_c = \mathbf{Z}^\top \mathbf{S}_s \mathbf{Z}, \qquad \mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{P} & \mathbf{B}_c \end{bmatrix} \in \mathbb{R}^{24\times 11}, \tag{10}$$

we formed an 11×11 block penalty $\mathbf{S} = \mathrm{blockdiag}(\mathbf{0}_{2\times 2}, \mathbf{S}_c)$. The intercept and Product B indicator were not penalized. One linear Northing direction also remains unpenalized inside the centered smooth. The observed Northing and Product covariates were identical at every Easting, so $\mathbf{X}$ and $\mathbf{S}$ were shared across all 199 response vectors within this manual construction.

4.2 Penalized least squares, GCV, and effective degrees of freedom

At each $e \in \mathcal{E}$ and for fixed $\lambda > 0$, the manual coefficient estimate solved

$$\widehat{\boldsymbol{\alpha}}_{e,\lambda} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{S})^{-1} \mathbf{X}^\top \boldsymbol{y}_e, \qquad \widehat{\boldsymbol{y}}_{e,\lambda} = \mathbf{H}_\lambda \boldsymbol{y}_e, \qquad \mathbf{H}_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{S})^{-1} \mathbf{X}^\top. \tag{11}$$

The second element of $\widehat{\boldsymbol{\alpha}}_{e,\lambda}$ is the unpenalized Product B coefficient, denoted $\widehat{\beta}_B(e;\lambda)$. Model complexity was measured by $\mathrm{EDF}(\lambda) = \mathrm{tr}(\mathbf{H}_\lambda)$, with smooth-specific EDF equal to total EDF minus the two separately modeled parametric columns. With the common design, $\mathbf{H}_\lambda$ and its trace are constant across Eastings at fixed λ, but the residual sum of squares and GCV depend on $\boldsymbol{y}_e$.

We computed the GCV score in equation (3), using $\mathrm{RSS}_e(\lambda) = \|\boldsymbol{y}_e - \mathbf{H}_\lambda \boldsymbol{y}_e\|_2^2$. The search used a grid of 400 equally spaced values of $\ell = \log(\lambda_{\mathrm{manual}})$ on $[-1, 23]$, i.e., $\lambda = \exp(\ell)$. Thus, $199 \times 400 = 79{,}600$ manual GCV values were evaluated. Logarithmic spacing covers weak through very strong penalization without treating the spacing of λ itself as uniform. The numerical value of $\lambda_{\mathrm{manual}}$ is specific to this basis and penalty; it cannot be directly equated with the `mgcv` smoothing parameter in the later package experiments.

4.3 Grid minima, local refinement, and candidate comparison

An interior grid point was flagged as a local-minimum candidate when its GCV score was lower than its immediate grid neighbors, subject to the implementation's smooth EDF rule $\mathrm{EDF}_{\mathrm{smooth}} > 1.01$ to avoid classifying effectively linear numerical plateaus as curvature-bearing minima. For a detected grid position, a parabola was fitted to the three neighboring sampled scores; its vertex was restricted to the neighboring grid interval and the corresponding spline fit was then recomputed. We refer to the resulting locations as *locally refined numerical candidates*. This three-point interpolation should not be mistaken for exhaustive continuous optimization, analytic stationarity testing, or proof of global optimality.

We also evaluated an exact limiting model as $\lambda \to \infty$. Since the curvature penalty leaves a linear Northing component unpenalized, the limiting design is

$$\mathbf{X}_\infty = \begin{bmatrix} \mathbf{1} & \mathbf{P} & \mathbf{N} - \overline{N}\mathbf{1} \end{bmatrix}, \quad \mathrm{EDF}_\infty = 3, \quad \mathrm{GCV}_{e,\infty} = \frac{24\,\mathrm{RSS}_{e,\infty}}{(24-3)^2}. \tag{12}$$

The infinite-penalty limit was calculated explicitly by ordinary least squares rather than approximated solely by the largest grid value. At each Easting, the lowest *listed* candidate was chosen from the locally refined interior minima, the exact linear limit, and both finite grid endpoints. The comparison of a best detected interior candidate with the exact linear limit was summarized by

$$\Delta_{\mathrm{struct}}(e) = \mathrm{GCV}_{e,\text{best interior}} - \mathrm{GCV}_{e,\infty}. \tag{13}$$

A negative gap indicates a better score for an evaluated interior candidate; a positive gap indicates a better score for the linear limit. This statistic is undefined when no qualifying interior candidate was detected. It differs from the optimizer gap defined for `mgcv` in the next phase: Stage 2 evaluates *candidate preference*, not the performance of an automatic optimization routine.

4.4 Reproducibility and boundaries of the manual experiment

The independent analysis used Base R matrix operations with explicit basis, penalty, and grid construction. Checks on the saved outputs included the B-spline partition of unity, the numerical penalty rank and centering condition, reproduction of the original three-slice GCV scores, and the identity between direct fixed-basis leave-one-out prediction errors and the leverage-based expression

$$\mathrm{LOOCV}_e(\lambda) = \frac{1}{24}\sum_{i=1}^{24}\left\{\frac{Y_i(e) - \widehat{Y}_i(e;\lambda)}{1 - h_{ii}(\lambda)}\right\}^2. \tag{14}$$

This leave-one-out criterion is a check on the fixed-basis linear-smoother calculation; it is not identical to GCV, which replaces individual leverage corrections with a trace-based adjustment. The source CSVs preserve all evaluated GCV values and coefficients, as well as the refined candidates and the candidate summaries. The analysis did not model dependence between Eastings or establish uncertainty in the estimated treatment trajectory.

4.5 Automatic GAM and fixed-grid comparisons (Stages 3A–3C)

Common package specification. The second computational experiment investigated the model used in the original software analysis, rather than numerically substituting the manually assembled spline. For each Easting e, we fitted the Gaussian additive model

$$Y_i(e) = \beta_0(e) + \beta_B(e)P_i + f_e(N_i) + \varepsilon_i(e), \qquad f_e \sim \texttt{s(Northing,bs="bs",k=10,m=c(3,2))}, \tag{15}$$

using `mgcv::gam()` with `method="GCV.Cp"`. Product E was the reference category, so $\widehat{\beta}_B$ is the B-minus-E adjusted contrast. The same formula, family, basis settings, and response observations were used in all automatic and fixed-smoothing fits. There was one penalized smooth and one package smoothing parameter, supplied to the fitting routine as `sp`. Accordingly, throughout the `mgcv` analyses we write $\lambda_{\mathrm{mgcv}} = \texttt{sp}$ and $\ell = \log(\lambda_{\mathrm{mgcv}})$. This parameter does not share a numerically interchangeable scale with the independent manual λ from Stage 2, because the respective basis and penalty constructions differ. The fitted *criterion scores* and coefficients at a given Easting can be compared as separate implementations, but identical finite parameter coordinates are not expected.

Stage 3A: selected-Easting pilot. We first studied Eastings 212, 214, 216, 218, 232, and 234. At each location, an automatic GCV fit supplied $\widehat{\lambda}_{e,\mathrm{auto}}$ and its score, effective degrees of freedom (EDF), and $\widehat{\beta}_B$. We then fitted the same model at each of 600 fixed values on

$$\ell_j = -12 + \frac{37(j-1)}{599}, \quad \lambda_j = \exp(\ell_j), \qquad j = 1, \ldots, 600. \tag{16}$$

The grid covers $[-12, 25]$ on the natural-log scale. It supplied $6 \times 600 = 3{,}600$ fixed-parameter scores for the pilot, including the GCV profile near the original three-slice observations. The automatically selected values were compared with the lowest evaluated grid scores. Where an automatic parameter lay beyond the finite grid's right edge, its score was retained as its own evaluated solution rather

than assigned to the endpoint.

Stage 3B: automatic selection and an exact known alternative. The same automatic GAM was then fitted separately at all 199 Eastings. In addition to its selected λ, GCV score, smooth EDF, total EDF, residual sum of squares, Product B coefficient, and convergence status, the experiment calculated the exact penalty-nullspace limit via the ordinary least-squares design in equation (12). The screen

$$D_{\text{linear}}(e) = \text{GCV}_{e,\text{auto}} - \text{GCV}_{e,\infty} \tag{17}$$

identifies an automatic fit as worse than the exact linear model when $D_{\text{linear}}(e) > 10^{-4}$. This comparison provides a *sufficient demonstration of an inferior automatic score relative to a known alternative*; its converse does not establish that the automatic solution is the lowest possible GCV. The automatic fits were also refitted at their recorded smoothing parameters as a calculation-consistency check. The study does not regard optimizer convergence as proof of global optimality.

Stage 3C: complete evaluated package criterion profiles. Finally, the 600-point grid in equation (16) was applied to every one of the 199 response vectors, producing $199 \times 600 = 119{,}400$ fixed-smoothing GAM evaluations. Each fit contributed the score $\text{GCV}_e(\lambda_j)$, smooth EDF, Product B coefficient, and RSS. An interior sampled minimum was detected at an index $j \in \{2, \ldots, 599\}$ when

$$\text{GCV}_e(\lambda_{j-1}) > \text{GCV}_e(\lambda_j) < \text{GCV}_e(\lambda_{j+1}), \qquad \text{EDF}_{\text{smooth}}(e, \lambda_j) > 1.01. \tag{18}$$

The rule is applied to the *fixed-grid* scores, not to analytic first derivatives. Unlike Stage 2's three-point vertex interpolation, these Stage 3C candidate records are at grid positions and were not refined within Stage 3C; the subsequent Stage 4A experiment refines these same identified candidates. The finite-grid best candidate was $\widehat{\lambda}_{e,\text{grid}} = \arg\min_{\lambda_j} \text{GCV}_e(\lambda_j)$; if this occurred at $\ell = 25$, we label it a *right-boundary grid solution*, not a proved infinite-penalty optimum. The exact linear-limit fit was always available as an independently computed benchmark.

4.6 Stage 4A: bracketed refinement and validation of package GCV minima

Stage 4A is a numerical validation of the finite-grid evidence, not a change in response data or GAM specification. For each of the 218 grid-detected interior minima indexed by j in equation (18), the original 600-point profile supplied the bracket $[\ell_{j-1}, \ell_{j+1}]$, and we applied one-dimensional bounded minimization to the identical fixed-smoothing-parameter `mgcv` GCV objective,

$$\widetilde{\ell}_{e,j} = \underset{\ell \in [\ell_{j-1}, \ell_{j+1}]}{\arg\min} \text{GCV}_e(e^{\ell}), \qquad \widetilde{\lambda}_{e,j} = e^{\widetilde{\ell}_{e,j}}. \tag{19}$$

The implementation used `stats::optimize()` with a requested tolerance of 10^{-9} in log-smoothing-parameter units. Each objective evaluation refitted the same Gaussian `mgcv` model at fixed `sp`; the GCV was recomputed as $n\,\text{RSS}/(n - \text{EDF})^2$ from the fit. Local checks compared the refined value with its grid center and nearby log-parameter values on either side; tolerances of 10^{-5} in GCV were used for numerical consistency. This procedure does not search for additional basins outside the

Table 2: Reference sets and contrasts used in the same-response, same-model numerical comparisons. Each reference score is paired with the Product B coefficient of the candidate attaining that score. None is asserted to be a mathematically global GCV minimum.

Reference or contrast	Definition and interpretation
Best grid (Stage 3C)	Lowest GCV among the 600 fixed `sp` values; includes neither a separately computed exact linear limit nor subsequent local refinement.
Exact linear limit	OLS model with intercept, Product B, and linear Northing; evaluated directly at the infinite-curvature-penalty limit.
Best alternative (Stage 4A)	Minimum GCV among best grid, exact linear limit, and all refined detected interior candidates; *excludes* the automatic fit.
Best evaluated (Stage 4A)	Lower GCV of the automatic fit and the best alternative; *includes* the automatic fit.
Grid contrast	$D_{\text{grid}} = \text{GCV}_{\text{auto}} - \text{GCV}_{\text{grid}}$ and $\Delta_{\beta,\text{grid}} = \widehat{\beta}_{B,\text{auto}} - \widehat{\beta}_{B,\text{grid}}$.
Stage 4A contrast	$D_{\text{alt}} = \text{GCV}_{\text{auto}} - \text{GCV}_{\text{alt}}$; inferior when $D_{\text{alt}} > 10^{-4}$. Corresponding coefficient difference uses the same *alternative*.
Linear reference contrast	$\widehat{\beta}_{B,\text{auto}} - \widehat{\beta}_{B,\infty}$; a different question from automatic-versus-alternative discrepancy.

detected brackets, and local refinement alone does not certify the global minimum.

As an independent reproduction gate, the analysis re-evaluated previously saved fixed-parameter fits at Eastings 214, 216, and 374 using the original processed Yield and compared both GCV and Product B to their Stage 3C records. For each Easting we then compared the best refined interior candidate with the exact linear limit and retained the original best grid candidate as an additional cross-check. The benchmark set also included the recorded automatic fit. The resulting alternative and best-evaluated scores are

$$\begin{aligned} \text{GCV}_{e,\text{alt}} &= \min\{\text{GCV}_{e,\infty}, \text{GCV}_{e,\text{grid}}, \text{GCV}_{e,\text{refined interior}}\}, \\ \text{GCV}_{e,\text{evaluated}} &= \min\{\text{GCV}_{e,\text{auto}}, \text{GCV}_{e,\text{alt}}\}. \end{aligned} \tag{20}$$

where the refined interior option is absent when no interior dip is detected. We use the term *inferior automatic selection* when $\text{GCV}_{e,\text{auto}} - \text{GCV}_{e,\text{alt}} > 10^{-4}$. The 10^{-4} threshold is numerical, not a hypothesis-test threshold. We report the corresponding signed Product B difference separately. The coefficient attached to a best-evaluated score is the coefficient of its attaining fit, not an average across candidate fits. Table 2 distinguishes the historical Stage 3C grid reference from the expanded Stage 4A reference sets. Stage 4A does not implement analytic stationary-point equations or a new global-search algorithm.

Score and coefficient diagnostics. The following contrasts separate the optimization question from the scientific coefficient question:

$$D_{\text{grid}}(e) = \text{GCV}_{e,\text{auto}} - \text{GCV}_{e,\text{best grid}}, \tag{21}$$

$$\Delta_{\beta,\text{grid}}(e) = \widehat{\beta}_{B,e,\text{auto}} - \widehat{\beta}_{B,e,\text{best grid}}, \tag{22}$$

$$D_{\text{struct,mgcv}}(e) = \text{GCV}_{e,\text{best detected interior}} - \text{GCV}_{e,\infty}\,. \tag{23}$$

For interpretation after Stage 4A, let $\widehat{\beta}_{B,e,\text{evaluated}}$ denote the coefficient of the lowest-scoring model in equation (20). There is an exact accounting identity,

$$\widehat{\beta}_{B,e,\text{auto}} - \widehat{\beta}_{B,e,\infty} = \underbrace{\left(\widehat{\beta}_{B,e,\text{auto}} - \widehat{\beta}_{B,e,\text{evaluated}}\right)}_{\text{automatic-versus-best evaluated contrast}} + \underbrace{\left(\widehat{\beta}_{B,e,\text{evaluated}} - \widehat{\beta}_{B,e,\infty}\right)}_{\text{best-evaluated-versus-linear contrast}} . \quad (24)$$

The identity separates the automatic-versus-best-evaluated contrast from the best-evaluated-versus-linear reference contrast. The first term is zero whenever the automatic fit is best evaluated, even if an alternative evaluated fit has a slightly different coefficient. The second term is a *model-comparison contrast*, not an isolated causal effect of curvature. The identity does not estimate bias or imply that the absolute values or medians of the separate terms add. When the linear limit is itself the best evaluated model, the second term vanishes.

A value $D_{\text{grid}}(e) > 10^{-4}$ defines an *inferior automatic selection relative to the evaluated grid* (Stage 3C); Stage 4A separately uses the best alternative, which can also be the exact linear limit or a locally refined interior candidate. This is a numerical comparison threshold, not a hypothesis-test cutoff. The structural contrast is undefined when the grid detector finds no qualifying interior minimum. A coefficient difference is a change *between fits on the same observed response*, not a measured bias relative to the unknown true treatment effect. The 199 Easting-indexed fits reuse the same pass structure and should not be interpreted as 199 statistically independent experiments.

Computational traceability. The pilot, automatic field-wide, fixed-grid, and local-refinement experiments saved separate CSV tables of selected fits, grid values, detected minima, refined minima, and comparative diagnostics. The accompanying project includes canonical input CSVs, the Stage 4A R script, the saved Stage 4A validation gate, and exhibit-generation scripts. Rebuilding manuscript graphics from these data does not refit the original GAMs. In contrast, reproducing Stage 4A from scratch requires the original processed `Example1SmoothHarvestPasses.csv`, which was used by the run but is not included in the present Overleaf archive. The completed computations do *not* include stationary-point root finding using Reiss–Ogden's h-functions, full-field REML comparisons, or a controlled simulation of coefficient bias. Those remain outside the evidential scope of Paper 1.

5 Full-Field Numerical Results

5.1 Full-field GCV criterion landscape

Exhibit provenance. Figures labeled F01–F13 in this draft are the uploaded vector-PDF graphics from the October 20 research-notes package, inserted without re-estimating the models. Their numerical sources are the saved Stage 2 and Stage 3A–3C CSV outputs; the shorter curated summary CSV files contain rounded presentation values and are not treated as independent model runs.

Figure 2 shows the empirical criterion landscape under the manual spline construction. Its left panel presents absolute $\text{GCV}_e(\exp \ell)$, whereas the right panel subtracts the exact linear-limit score of the same Easting: $\text{GCV}_e(\exp \ell) - \text{GCV}_{e,\infty}$. The subtraction is constant within any one Easting and therefore does not change the locations of within-Easting extrema. It does, however, make the

relative preference between a finite-smoothing fit and the exact linear adjustment directly visible: negative relative values favor the finite-smoothing candidate according to GCV. Differences in absolute criterion levels among Eastings can reflect differences in residual variation; a high absolute score alone does not identify an optimizer failure.

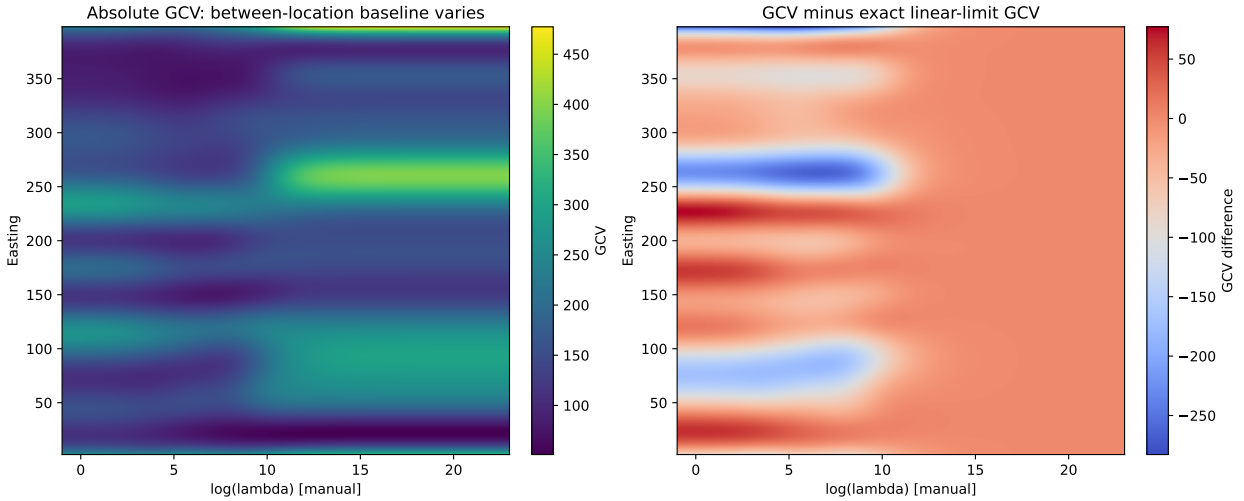


Figure 2: Independent manual spline GCV surfaces. Left: absolute GCV; right: the same score minus the exact linear-limit GCV within each Easting. The horizontal coordinate is $\log(\lambda_{\text{manual}})$ (400 fixed values) and cannot be compared numerically to the package smoothing parameter. Uploaded Figure F01. Original data: `Manual_Full_Field_Grid.csv` and `Manual_Full_Field_Summary.csv`.

5.2 Interior minima across the field

The finite-grid procedure detected at least one qualifying interior minimum at 189 of the 199 Eastings and no qualifying interior minimum at ten Eastings (Table 3). Among those with an interior minimum, 162 Eastings had one, 26 had two, and one Easting had three; hence **27 Eastings displayed multiple grid-detected interior minima**. Counting individual detected minima rather than locations yields 217 locally refined interior candidates. These are characteristics of the evaluated numerical criterion under the specified detector, not estimates of the number of all theoretical stationary points on $(0, \infty)$.

Figure 3 displays the spatial distribution of those counts. Multiple-minimum profiles occur in several clusters rather than solely near the original Eastings 214–218. Their existence establishes competing local score valleys under the manual construction; it does not establish that automatic `mgcv` fitting will choose an inferior one, because no automatic optimizer was used in Stage 2.

Table 3: Distribution of grid-detected interior GCV minima across the 199 Eastings under the independent manual spline construction. The count excludes the finite grid endpoints and the exact linear-limit candidate. Source: `Manual_Full_Field_Summary.csv`, column `Interior_Dip_Count`.

Detected interior minima	Eastings	Percent of Eastings
0	10	5.0%
1	162	81.4%
2	26	13.1%
3	1	0.5%
Total	**199**	**100.0%**

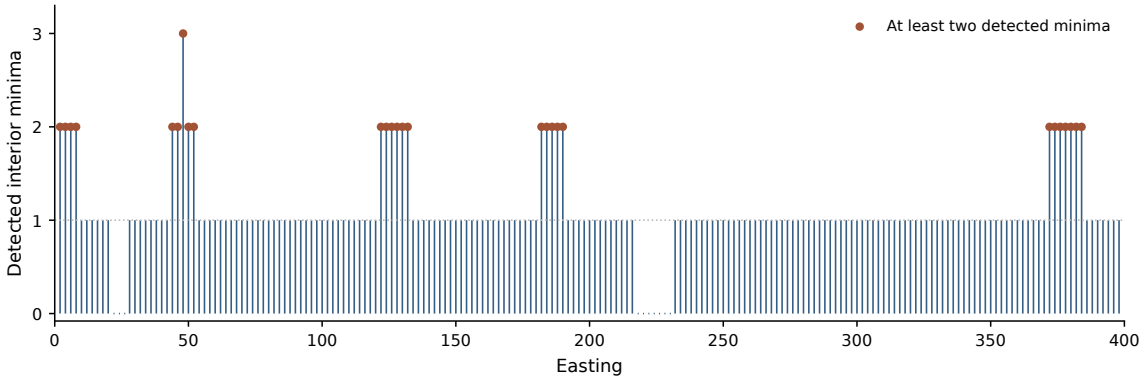


Figure 3: Count of detected interior GCV minima by Easting in the manual 400-point log-smoothing grid. Eastings with two or three minima are highlighted. The rule requires a sampled local turning point and smooth EDF greater than 1.01. Source: `Manual_Full_Field_Summary.csv`, `Interior_Dip_Count`; individual candidate records are in `Manual_Candidate_Minima.csv`.

5.3 The interior-versus-linear comparison

Comparing the locally refined minima against the exact linear limit, the lowest listed candidate was an interior minimum at 181 Eastings and the linear-limit fit at 18. Figure 4 depicts the structural score gap in equation (13); Figure 5 separately maps the lowest-scoring candidate regime. A negative gap indicates that an interior candidate improves GCV relative to the straight-line adjustment, whereas a positive gap means that the linear limit has the lower score. At the ten locations with no detected interior minimum the structural gap is left undefined, rather than imputed as zero.

Table 4: Adjacent observed Eastings with different lowest-listed manual smoothing regimes. These are discrete candidate classifications, not estimated locations of continuous spatial discontinuities. Source: `Manual_Full_Field_Summary.csv`, sorted by `Easting` and compared using `Lowest_Candidate`.

Adjacent Eastings	Earlier candidate	Later candidate
20–22	Interior	Exact linear limit
30–32	Exact linear limit	Interior
170–172	Interior	Exact linear limit
176–178	Exact linear limit	Interior
212–214	Interior	Exact linear limit
232–234	Exact linear limit	Interior

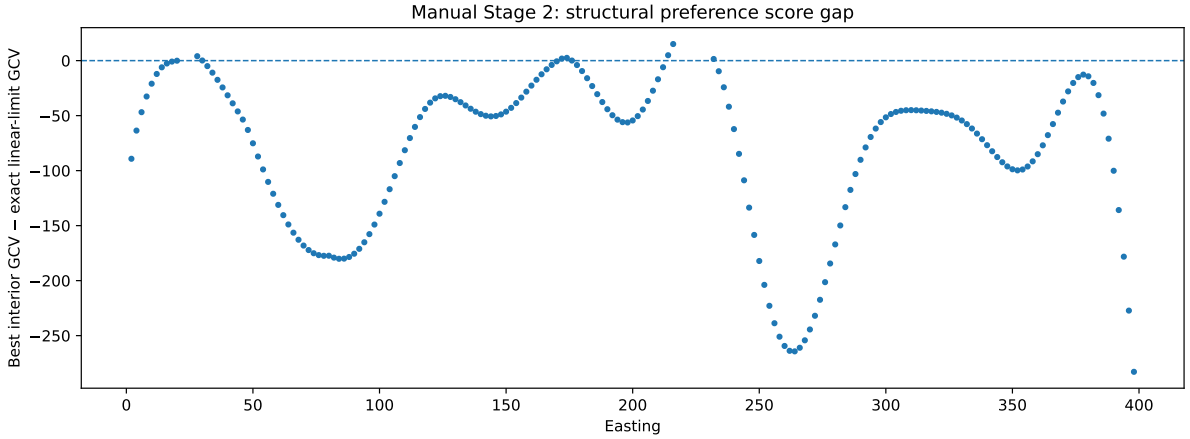


Figure 4: Manual Stage 2 structural criterion gap: GCV at the best detected interior minimum minus the exact linear-limit GCV at each Easting. Negative values favor an interior candidate; positive values favor the linear limit. The gap is undefined for Eastings without a qualifying detected interior minimum. Uploaded Figure F08. Original data: `Manual_Full_Field_Summary.csv`.

The observed candidate preference changed between six pairs of adjacent Eastings, listed in Table 4. These are changes in the preferred *evaluated candidate* at the sampled Eastings, not demonstrations of continuous spatial discontinuities or of a mathematical crossing at a precisely located Easting. In particular, smoothing choice can shift between two regimes even if the underlying response vectors vary without a literal jump.

5.4 Revisiting the original three Eastings

Figure 6 and Table 5 recover the behavior that motivated the field-wide investigation. At Easting 214, the locally refined interior GCV was approximately 158.422, exceeding the exact linear-limit score of 153.498. At Easting 216, the corresponding scores were approximately 171.417 and 156.334. Easting 218 had no qualifying interior dip, and its exact linear-limit score was approximately 160.229. Therefore, all three original Eastings favored the exact linear limit among the listed manual candidates, even though two of them contained an interior local minimum. This example makes clear why locating an interior dip and identifying the lowest evaluated score are not equivalent tasks.

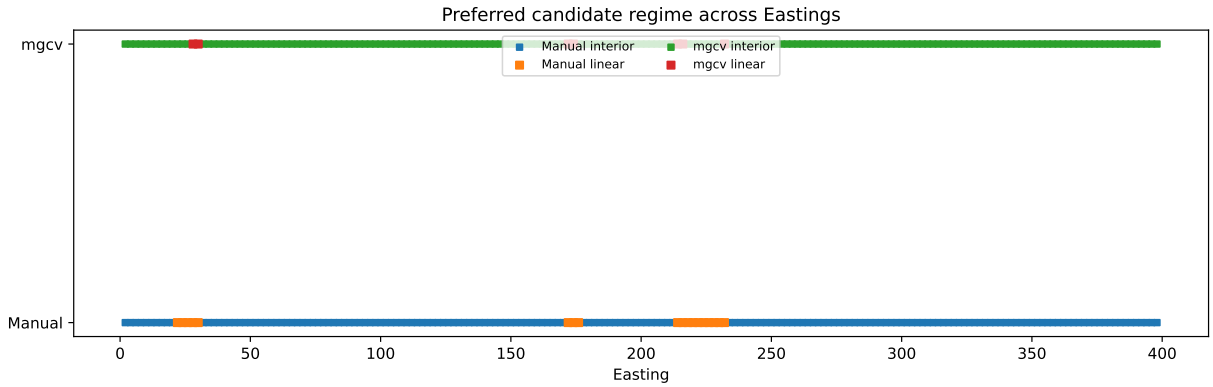


Figure 5: Preferred evaluated candidate regime across Eastings under the independent manual spline and the fixed-parameter `mgcv` model. This is a comparison of criterion preferences, not a map of optimizer failures or evidence of a discontinuous true treatment response. The near-tie at Easting 176 produces slightly different transition labels. Uploaded Figure F10. Original data: `Manual_Full_Field_Summary.csv` and `Stage3C_Auto_vs_Grid.csv`.

Table 5: Manual GCV candidates at the original Eastings. Interior GCV reports the smallest locally refined candidate, when one was detected; the exact linear-limit candidate is obtained by an OLS fit with a linear Northing term. All three Eastings favor the linear-limit score over the listed interior candidates. Source: `Manual_Full_Field_Summary.csv` and `Manual_Candidate_Minima.csv`.

Easting	Interior minima	Best interior GCV	Linear GCV	Structural gap	Linear $\widehat{\beta}_B$
214	1	158.422	153.498	+4.924	11.755
216	1	171.417	156.334	+15.082	11.467
218	0	—	160.229	—	11.190

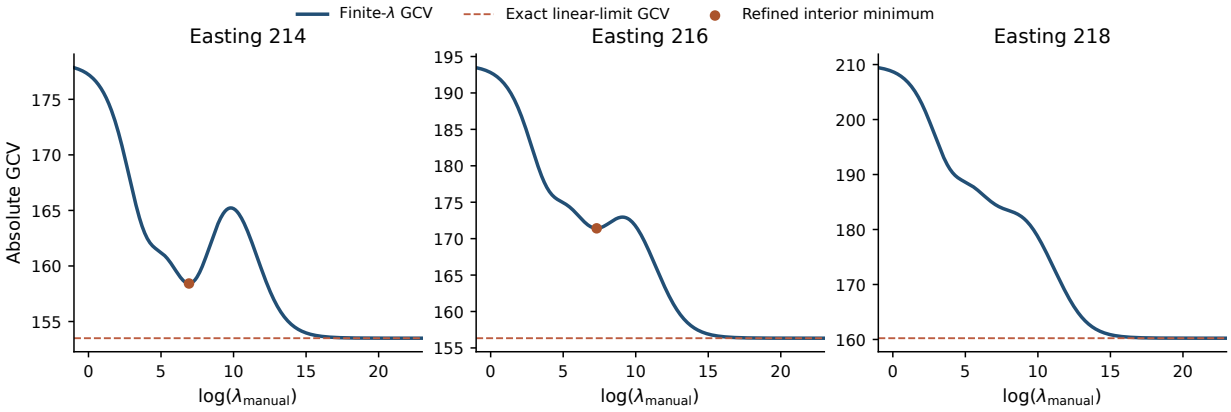


Figure 6: Manual Stage 2 absolute GCV profiles at Eastings 214, 216, and 218. A dashed horizontal line is the exact linear-limit score. Dots identify locally refined interior GCV minima when detected. The horizontal smoothing-parameter coordinate belongs to the manual spline, not the `mgcv` parameterization. Sources: `Manual_Full_Field_Grid.csv`, `Manual_Candidate_Minima.csv`, and `Manual_Full_Field_Summary.csv`.

5.5 Treatment-coefficient consequences of manual smoothing choice

Because the unpenalized Product B coefficient is estimated jointly with the Northing smooth, the estimated contrast $\widehat{\beta}_B(e;\lambda)$ may differ between the selected interior candidate and the exact linear adjustment. Figure 7 displays the two resulting coefficient trajectories. Where the linear limit is the lowest listed candidate, the trajectories coincide by construction; elsewhere, departures reflect distinct smoothing choices under the same supplied Yield observations. These differences are numerical selection sensitivity, not measured bias relative to an unknown treatment-effect function.

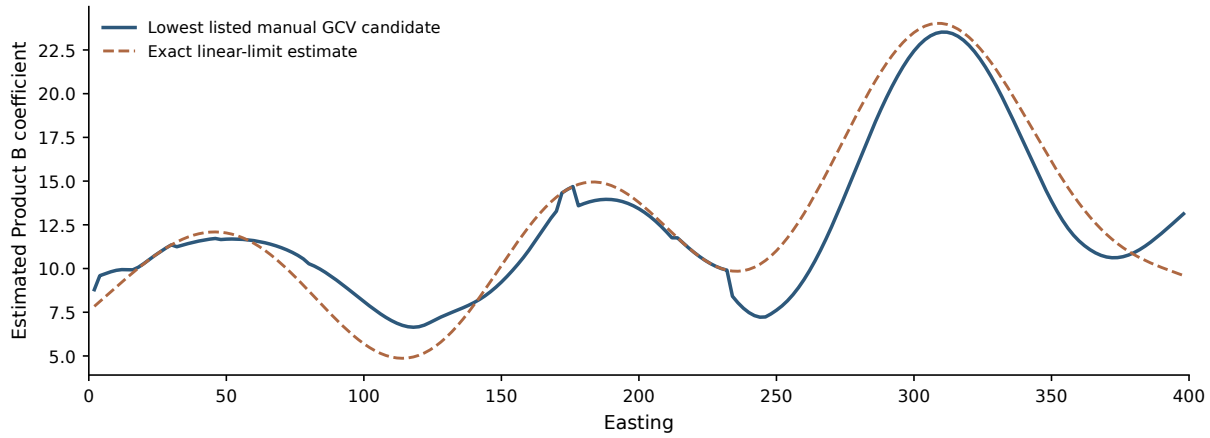


Figure 7: Manual Stage 2 Product B coefficient trajectories across the 199 Eastings. Solid line: coefficient from the lowest listed GCV candidate (locally refined interior, finite boundary, or exact linear); dashed line: coefficient from the exact linear-limit model at every Easting. Source: `Manual_Full_Field_Summary.csv`, columns `Lowest_Candidate_Product_B` and `Product_B_Linear_Limit`.

The manual experiment establishes a reproducible numerical picture of candidate multiplicity, regime preference, and coefficient sensitivity under a common design. It does not determine whether the software's automatic smoothing-parameter optimizer finds the lowest score. The separately parameterized package experiments in Stages 3A–3C address that subsequent question, and their finite-smoothing log-parameter coordinates should not be overlaid directly on those of the manual spline.

5.6 Six-Easting package pilot (Stage 3A)

The pilot directly compared automatic smoothing selection with the fixed-parameter GCV profile at six Eastings (Table 6; Figure 8). At Easting 214, the automatic score was 158.473 and the lowest evaluated grid score was 153.498, a difference of +4.975. At 216, the corresponding scores were 171.450 and 156.334, yielding +15.116. In both cases the lowest grid score occurred at the right edge $\ell = 25$, where the fitted Northing smooth was effectively linear. The associated automatic-minus-grid Product B differences were −0.358 and −0.418. Easting 218, by contrast, was automatically fitted very close to the linear limit; the automatic and right-edge grid scores were indistinguishable at the displayed precision. The automatic log smoothing parameter at 218 exceeded the grid boundary, so

Table 6: Stage 3A pilot: automatic GCV selection versus a fixed 600-point smoothing-parameter grid. The displayed differences are automatic minus lowest grid. Small negative gaps reflect the finite spacing of the grid rather than a failure of the automatic fit. Source: `Stage3A_Auto_vs_Grid.csv`.

Easting	Automatic GCV	Lowest-grid GCV	GCV gap	Product B gap
212	145.664	145.666	-0.002	+0.003
214	158.473	153.498	+4.975	-0.358
216	171.450	156.334	+15.116	-0.418
218	160.229	160.229	-0.000	-0.000
232	223.937	223.937	-0.000	-0.000
234	228.609	228.610	-0.000	-0.002

the apparent negligible negative gap reflects a more extreme automatic value than the largest finite grid point.

At Eastings 212, 232, and 234, the automatic selection agreed closely with the lowest evaluated grid candidate. In particular, the selected Product B coefficients changed from 9.906 at Easting 232 to 8.415 at 234 while the automatic and grid minima agreed to small numerical differences at each location. Thus a change in the preferred smoothing regime need not be accompanied by a detected optimizer shortfall. Such a change in coefficient is *not by itself proof of a spurious spatial discontinuity.*

5.7 Automatic selection over the full field (Stage 3B)

All 199 automatic `mgcv` fits completed and reported convergence. The exact linear-limit comparison nevertheless identified six locations with $D_{\text{linear}}(e) > 10^{-4}$: Eastings 28, 30, 172, 174, 214, and 216. At those locations, the returned automatic fit had a higher GCV score than a known, explicitly evaluated linear-nullspace alternative. The two largest such gaps occurred at Eastings 214 and 216, which had already been identified in the pilot. The remaining 193 locations passed this *single-alternative screen*, but the screen alone cannot establish the adequacy of their automatically selected solutions.

Figure 9 separates the spatial Product B estimates from the automatic-versus-linear score contrast. The red points in the lower panel identify the six cases where the automatic score is higher. The upper panel shows that the automatic and exact-linear coefficient trajectories are broadly similar but can differ meaningfully where a nonlinear Northing adjustment is retained. Such differences do not imply that the larger or smaller coefficient is closer to the unknown effect. Figure 10 shows the corresponding selected log-parameter and smooth EDF. Eleven automatic fits had smooth EDF within 0.01 of the linear nullspace value 1, specifically at Eastings 22–26 and 218–232. The abrupt changes in effective smoothing complexity at these locations motivate comparison with the finite-grid landscape, rather than independent interpretation of the log-smoothing trajectory as a measure of physical field variation.

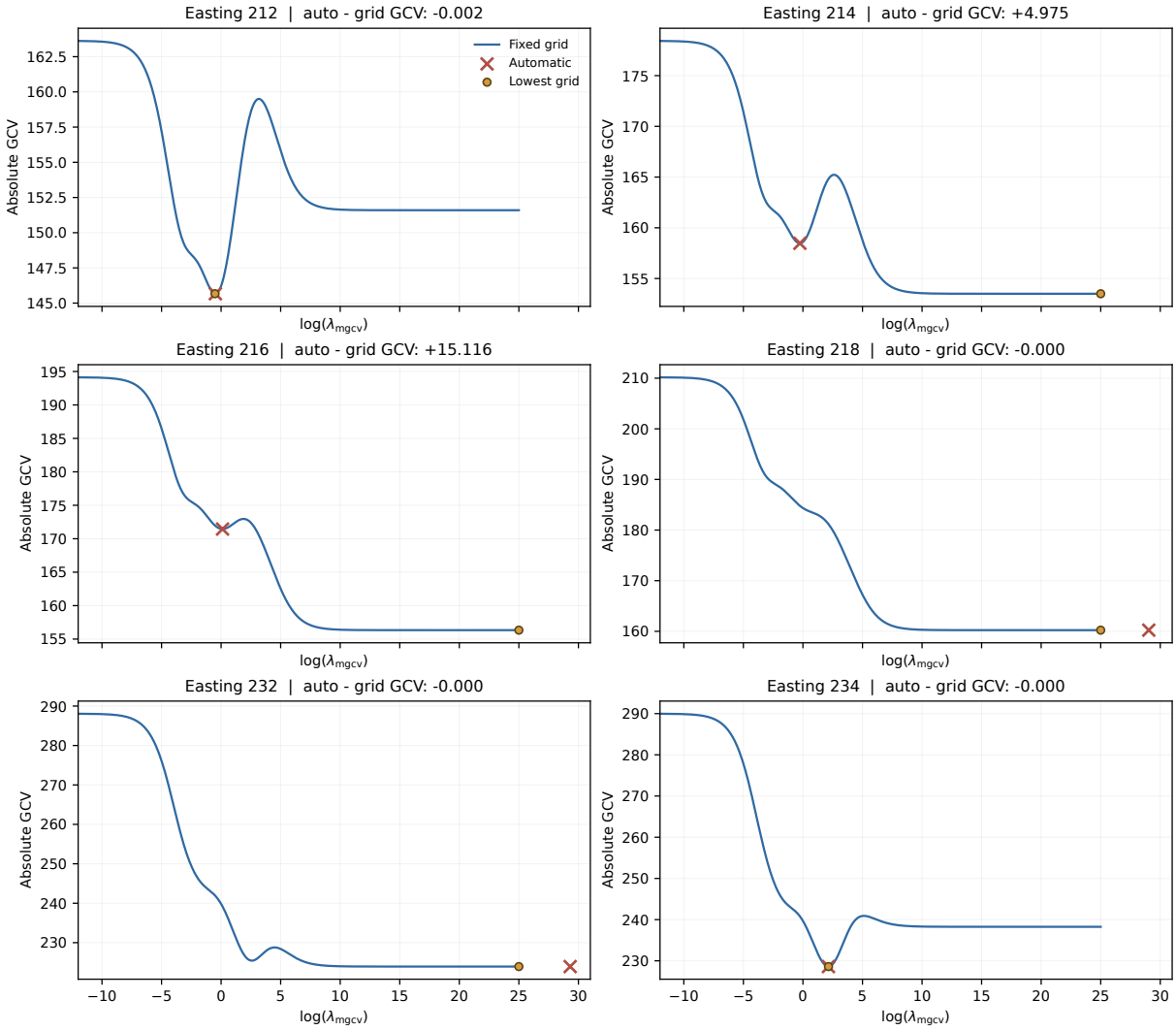


Figure 8: Six Stage 3A automatic-versus-fixed GCV profiles. Curves show 600 fixed λ_{mgcv} values per Easting; crosses identify automatic returns and filled dots show the lowest finite-grid score. The automatic smoothing parameter at Eastings 218 and 232 lies beyond the grid's right boundary and is therefore plotted separately rather than forced onto the last grid point. Source: `Stage3A_Fixed_Grid.csv` and `Stage3A_Auto_vs_Grid.csv`.

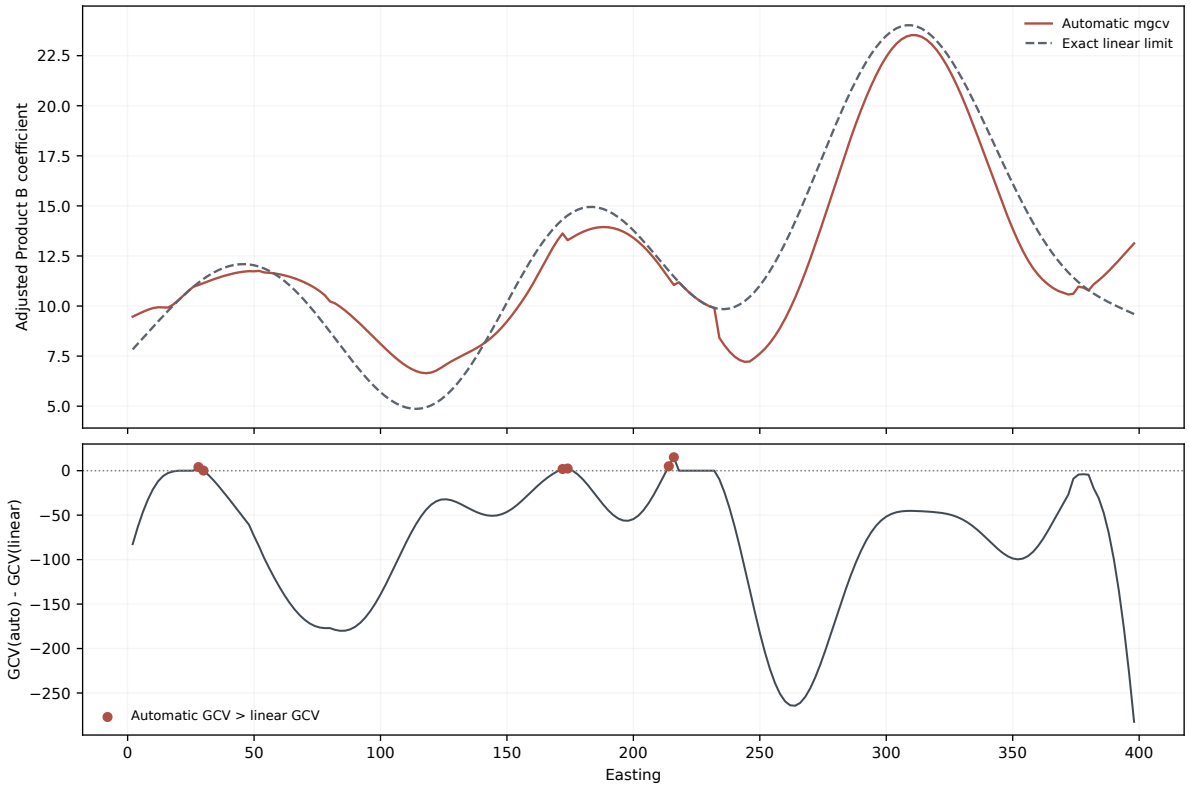


Figure 9: Stage 3B full-field automatic selection compared with the exact linear-limit fit. Upper panel: adjusted Product B coefficients. Lower panel: $\mathrm{GCV}_{\mathrm{auto}} - \mathrm{GCV}_{\mathrm{linear}}$, with positive red markers denoting the six inferior automatic scores relative to that known candidate. These are not yet automatic-minus-best-grid differences. Source: `Stage3B_Linear_Screen_and_Trajectories.csv`.

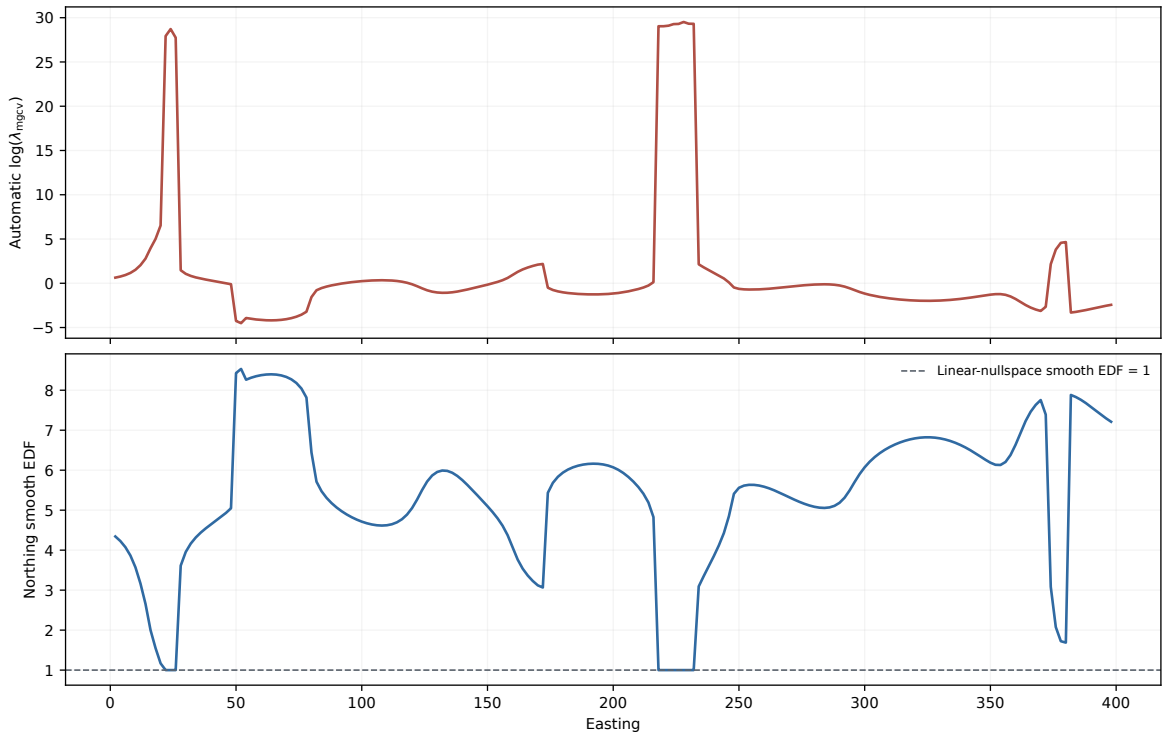


Figure 10: Stage 3B automatic log-smoothing parameter $\log(\lambda_{\mathrm{mgcv}})$ and the corresponding Northing-smooth EDF across Eastings. Smooth EDF approaching 1 indicates a fitted Northing adjustment close to the linear penalty-nullspace limit. Source: `Stage3B_Automatic_Selections.csv` and `Stage3B_Linear_Screen_and_Trajectories.csv`.

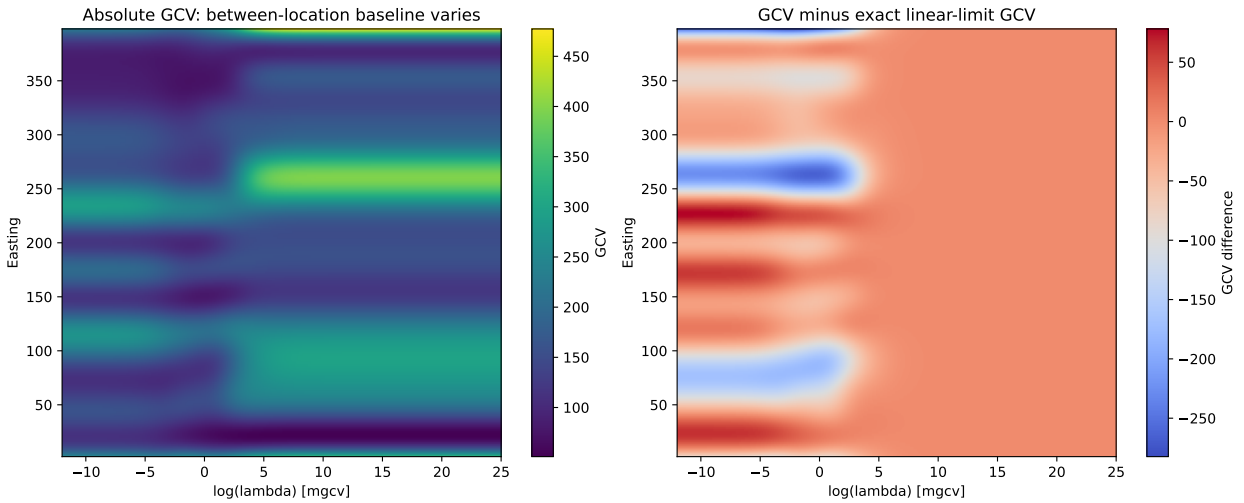


Figure 11: Package-based fixed-parameter GCV surfaces. Left: absolute GCV; right: the same GCV minus the exact linear-limit score within each Easting. The horizontal coordinate is $\log(\lambda_{\text{mgcv}})$ over 600 values and is not on the manual penalty scale in Figure 2. Uploaded Figure F02. Original data: `Stage3C_Full_Grid.csv` and `Stage3C_Auto_vs_Grid.csv`.

5.8 Full-field package GCV landscapes and multiple minima (Stage 3C)

Evaluating the same 600-point grid over all 199 Eastings yielded 119,400 GCV scores. Figure 11 presents two views of the resulting surface. The left panel shows the absolute score, whose overall level varies across Eastings. The right panel subtracts each Easting's exact linear-limit score from its entire fixed-parameter profile. This is a *row-specific constant translation*: it changes the displayed vertical origin of each GCV profile without changing its internal turning points. Negative values in the right panel indicate a finite smoothing parameter with GCV below the linear-limit benchmark, while positive values indicate a higher score.

The grid detector found at least one qualifying interior minimum at 189 Eastings and more than one at 28 Eastings. Specifically, 10, 161, 27, and 1 locations had zero, one, two, and three detected interior minima, respectively (Table 7). The 218 individual minima records came from the sampled grid, not analytic stationarity equations. The independently constructed manual spline had 27 multiple-minimum Eastings; the package grid had one additional such location, Easting 80. The two experiments employ different spline constructions and numerical candidate rules; this one-location discrepancy is not evidence that the original Yield values disagree. Figure 12 shows both distributions and where the multiple-minimum package profiles occur.

5.9 Inferior automatic selections relative to evaluated finite-grid alternatives

The complete grid comparison identified 16 Eastings for which the automatic GCV exceeded the lowest evaluated grid score by more than 10^{-4} (Table 8). These were 2, 28, 30, 48, 50, 52, 80, 172, 174, 214, 216, 372, 374, 376, 378, and 380. Six of the 16 were already worse than the exact linear limit in Stage 3B; the other ten required a different *finite* smoothing value to reveal the lower-scoring solution. At Easting 48 the package grid contained three interior minima, while some other inferior

Table 7: Distribution of grid-detected interior GCV minima across 199 Eastings. The manual spline used 400 log-smoothing values and local three-point refinement; the `mgcv` experiment used 600 fixed values and counted grid turning points. Their bases and penalty scales differ. Sources: `Manual_Full_Field_Summary.csv` and `Stage3C_Auto_vs_Grid.csv`.

Detected minima	Manual Stage 2	`mgcv` Stage 3C
0	10	10
1	162	161
2	26	27
3	1	1
Total Eastings	199	199
Eastings with multiple minima	27	28

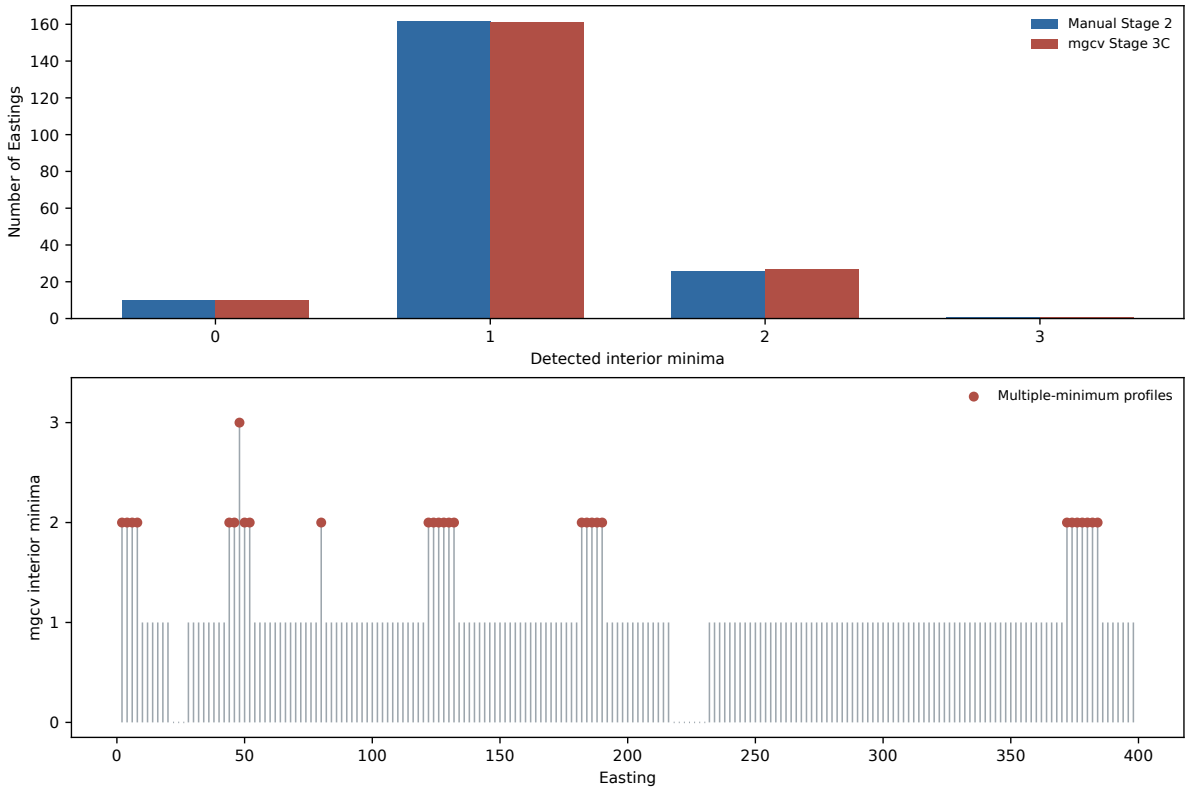


Figure 12: Upper panel: frequency of zero to three detected interior minima under manual Stage 2 and package Stage 3C spline specifications. Lower panel: number of package-based interior grid minima at each observed Easting, with multiple-minimum locations highlighted. Source: `Manual_Full_Field_Summary.csv`, `Stage3C_Auto_vs_Grid.csv`, and `Stage3C_Interior_Grid_Dips.csv`.

Table 8: All Eastings where the automatic `mgcv` score exceeded the lowest score on the 600-point fixed grid by more than 10^{-4}. Differences equal automatic minus lowest grid. The final column counts qualifying interior grid minima, not exact theoretical stationary points. Source: `Stage3C_Auto_vs_Grid.csv`.

Easting	GCV gap	Product B coefficient gap	Interior minima
2	+6.581	+0.686	2
28	+4.049	-0.117	1
30	+0.135	-0.205	1
48	+1.675	+0.089	3
50	+0.961	+0.060	2
52	+1.244	+0.076	2
80	+0.017	-0.017	2
172	+1.868	-0.694	1
174	+2.439	-1.237	1
214	+4.975	-0.358	1
216	+15.116	-0.418	1
372	+0.587	-0.045	2
374	+10.761	-0.014	2
376	+10.050	+0.291	2
378	+8.169	+0.141	2
380	+9.132	-0.131	2

automatic fits occurred at locations with only one detected interior minimum and a better-scoring high-smoothing alternative. Thus, neither multiple minima nor failure of the linear-limit screen is necessary for a positive optimizer gap under our numerical definitions.

The largest observed grid-comparison gap occurred at Easting 216, where $D_{\text{grid}} \approx 15.116$. In contrast, the automatic score at Easting 374 exceeded the lowest evaluated grid score by 10.761, but the favored alternative was a more flexible *finite* smoothing solution rather than a nearly linear one. Seventeen Eastings had their lowest fixed-grid score at the right boundary $\ell = 25$; the boundary record must not be called a mathematically proved global or exact infinite-penalty optimum. Figure 14 shows the spatial distribution of score gaps alongside the corresponding Product B coefficient differences.

5.10 Treatment-coefficient sensitivity and contrasting GCV regimes

Figure 15 compares the automatic, lowest-grid, and exact-linear Product B estimates. The automatic and lowest-grid trajectories often nearly coincide, but they separate at specific Eastings where different smoothing parameters are selected. The exact-linear coefficient trajectory provides a common reference for the consequences of flexible Northing adjustment; it is not presumed to be an unbiased truth. An especially useful contrast is between Eastings 174 and 374 (Table 9). At 174, a GCV disadvantage of only 2.439 corresponds to an automatic-minus-grid coefficient difference of -1.237. At 374, a substantially larger criterion shortfall, 10.761, corresponds to a difference of approximately -0.014. These are direct numerical counterexamples to interpreting the GCV shortfall as a proxy for the magnitude of coefficient sensitivity.

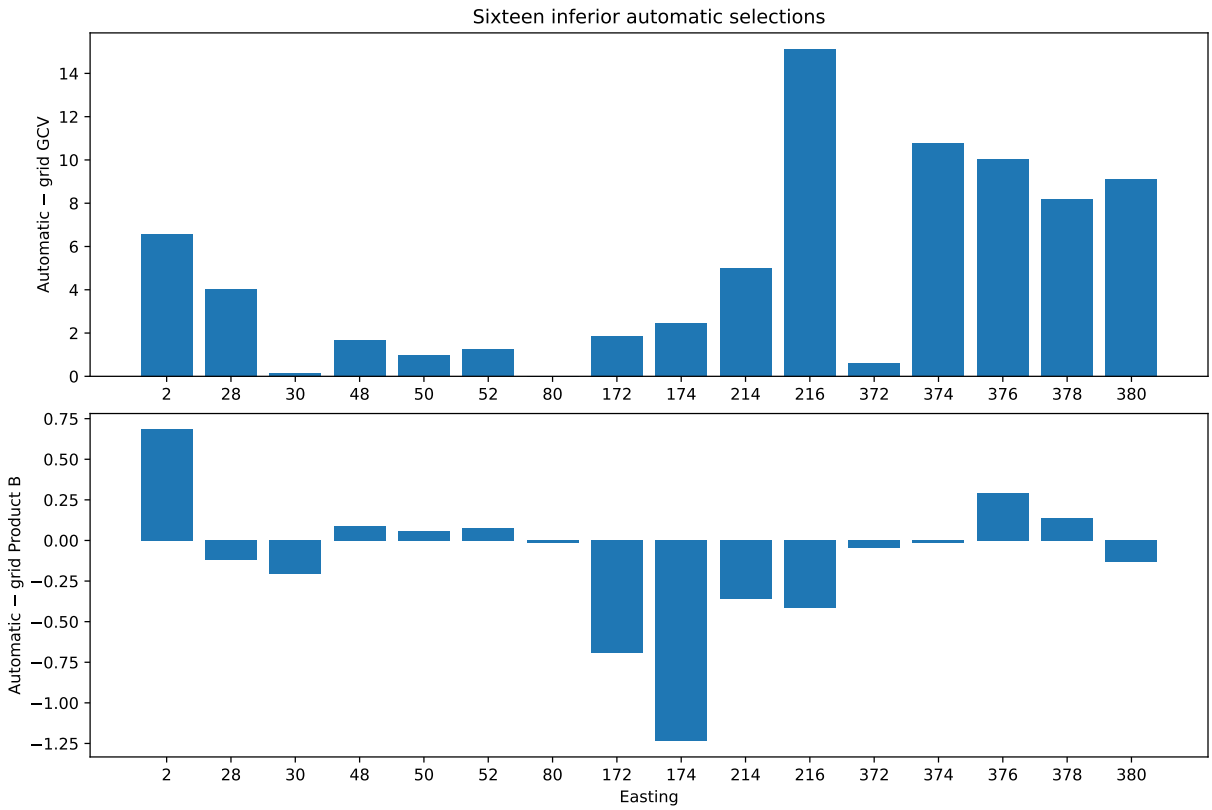


Figure 13: The sixteen observed Eastings with automatic GCV more than 10^{-4} above the lowest evaluated fixed-grid score. Upper bars show the criterion gaps; lower bars show automatic-minus-grid Product B differences. The signed lower values do not quantify estimation bias. Uploaded Figure F11. Original data: `Stage3C_Auto_vs_Grid.csv`; rounded companion: `sixteen_inferior_automatic_selections.csv`.

Table 9: Contrasting score and treatment-coefficient sensitivity. A larger GCV shortfall need not correspond to a larger change in the unpenalized Product B estimate. Differences equal automatic minus lowest evaluated grid. Source: `Stage3C_Auto_vs_Grid.csv`.

Easting	GCV gap	Product B gap	Interior minima
174	+2.439	-1.237	1
374	+10.761	-0.014	2

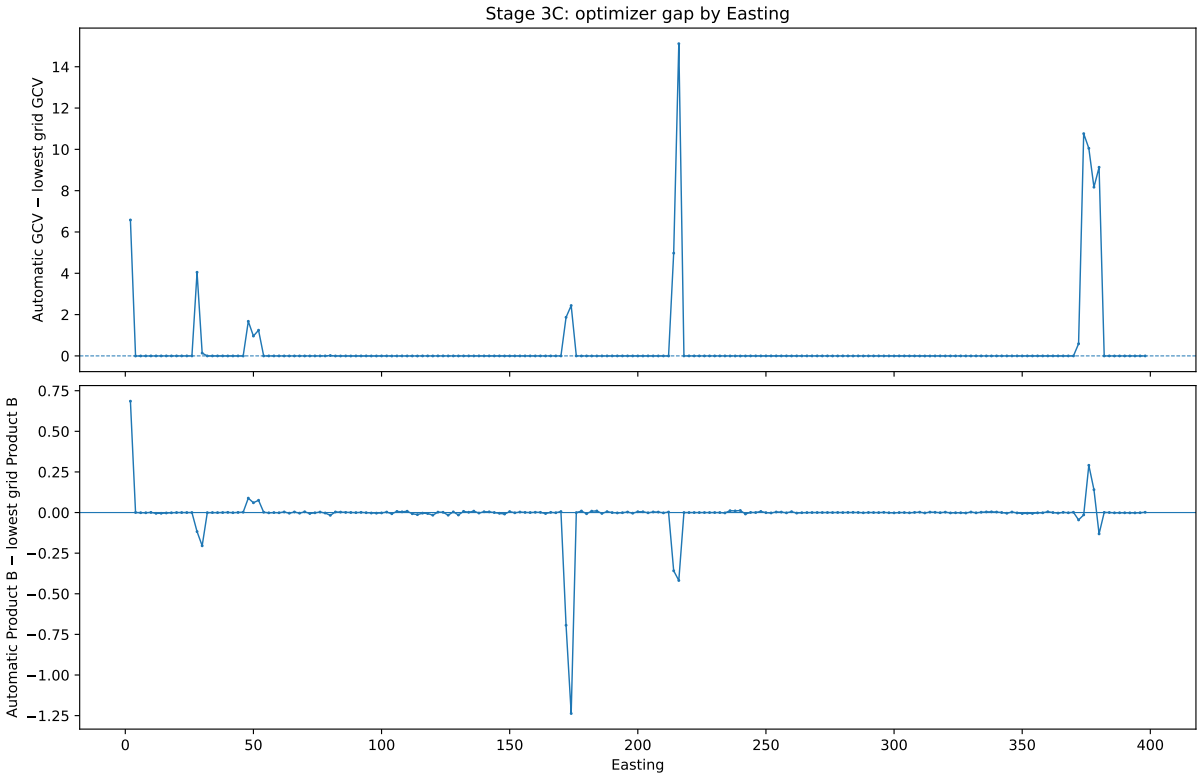


Figure 14: Full-field automatic-minus-lowest-grid GCV gaps (upper panel) and adjusted Product B coefficient differences (lower panel). The panels have different units; their magnitudes are not interchangeable. Uploaded Figure F07. Original data: `Stage3C_Auto_vs_Grid.csv`.

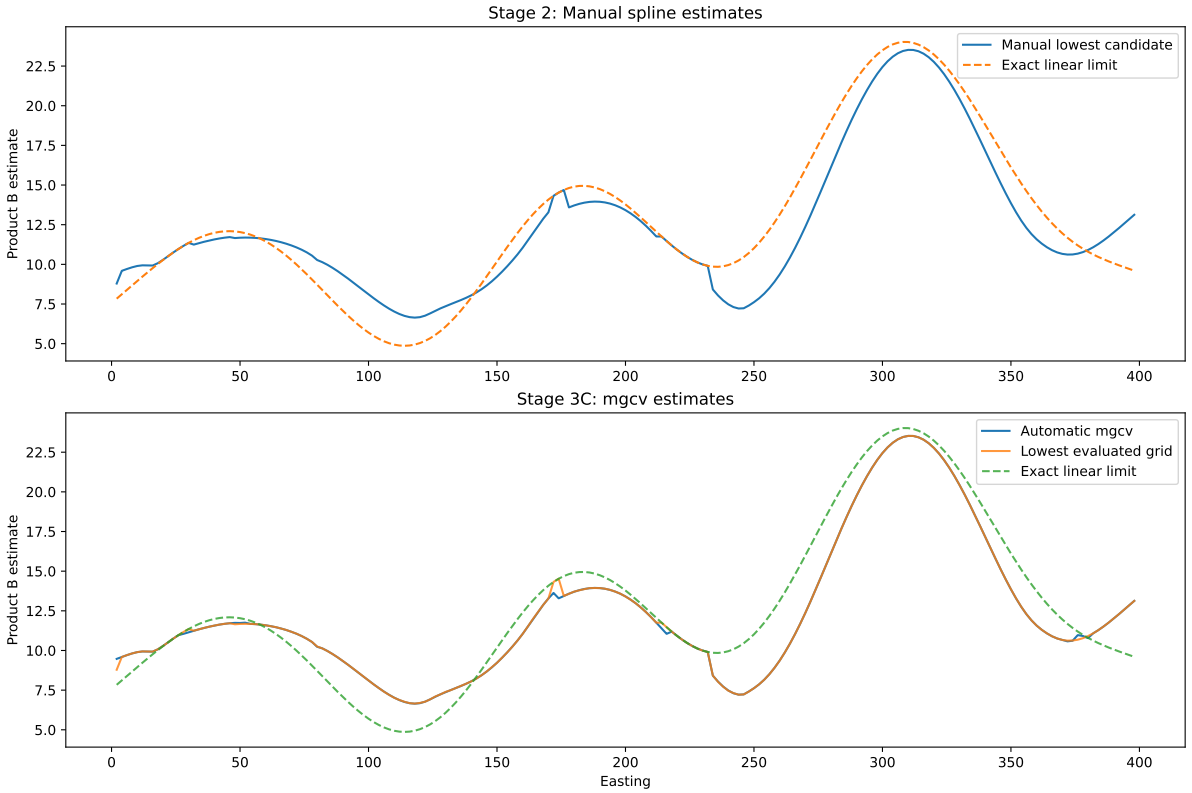


Figure 15: Two panels separating smoothing constructions. Top: manual lowest-listed candidate versus exact linear-limit Product B coefficients. Bottom: automatic `mgcv`, lowest evaluated 600-point `mgcv` grid, and exact linear-limit Product B estimates. Finite-smoothing penalty scales differ; the linear-limit fits agree. Uploaded Figure F06. Original data: `Manual_Full_Field_Summary.csv` and `Stage3C_Auto_vs_Grid.csv`.

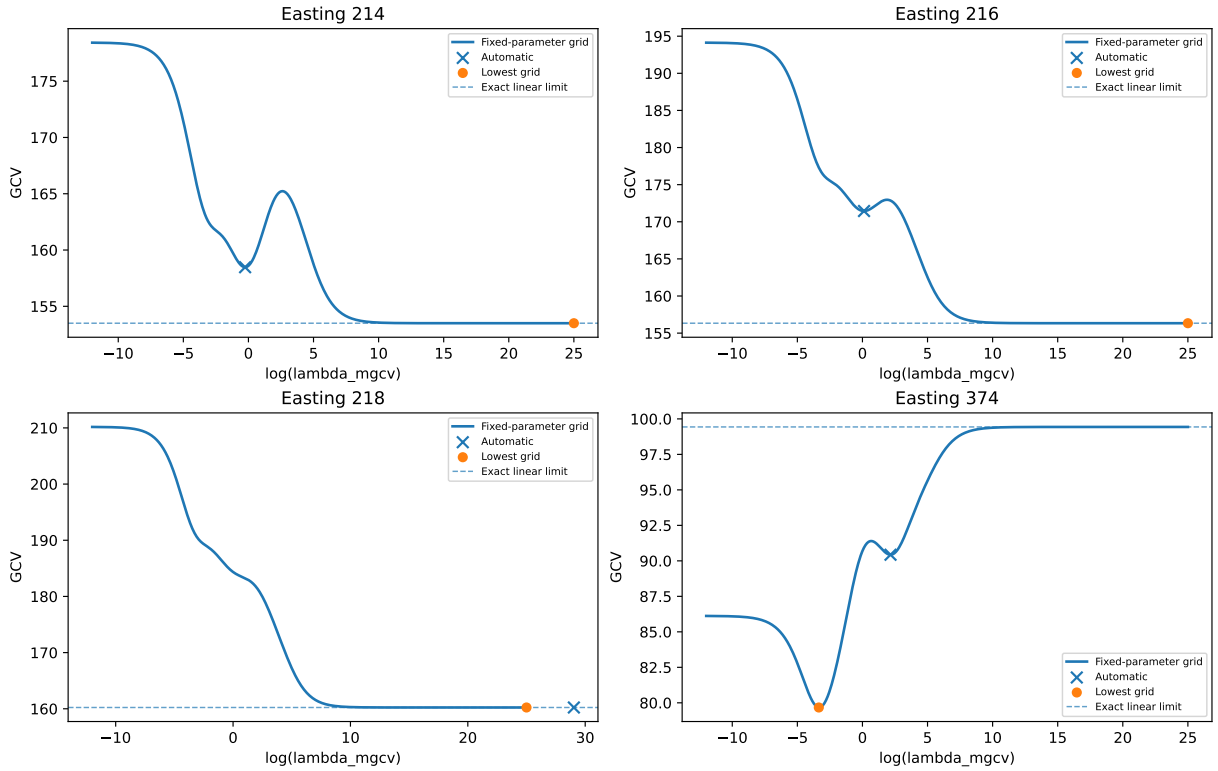


Figure 16: Four `mgcv` fixed-grid GCV profiles (Eastings 214, 216, 218, and 374). The automatic choice, lowest evaluated grid point, and exact linear-limit level are distinguished within each panel. All panels plot $\log(\lambda_{\text{mgcv}})$ but use their own absolute-GCV vertical scales. Uploaded Figure F03. Original data: `Stage3C_Full_Grid.csv` and `Stage3C_Auto_vs_Grid.csv`.

Figure 16 makes the selection mechanisms explicit. At Easting 216, the automatic solution lies near an interior GCV valley despite a lower score at the strongly penalized, nearly linear end of the evaluated domain. At Easting 374, the GCV profile contains two distinct detected interior minima. The first, near $\log(\lambda_{\text{mgcv}}) = -3.352$, had GCV approximately 79.671 and smooth EDF about 7.911, whereas the automatic fit stopped near $\log(\lambda_{\text{mgcv}}) = 2.157$, with GCV about 90.432 and smooth EDF about 3.081. The lower-GCV candidate at 374 was therefore *more flexible*, not more strongly penalized. Increasing the smoothing parameter beyond the automatic solution ultimately raised GCV toward the linear-limit score of approximately 99.434. Stronger penalization reduces EDF but can increase residual error enough to worsen GCV; neither GCV nor the treatment coefficient is required to change monotonically with smoothing strength.

Within-Easting criterion geometry at Easting 374. Figure 17 isolates the two sampled local minima in the package GCV profile. The first, near $\log \lambda_{\text{mgcv}} = -3.352$, scored 79.671; the other, near 2.145, scored 90.432. The automatic fit selected near the second basin, even though the first scored lower. The difference in GCV is a within-Easting comparison of two smoothing choices, not a comparison of the Yield scores at two different spatial locations. Table 10 reports the associated EDF and Product B values, with the automatic fit distinguished from the sampled grid minimum.

Table 10: Two sampled interior minima and the automatic solution at Easting 374. The automatic value is from a separate automatic fit, not an additional fixed-grid turning point. The score and coefficient values are rounded presentation figures. Source: `Stage3C_Interior_Grid_Dips.csv` and `Stage3C_Auto_vs_Grid.csv`; curated summary `easting_374_stationary_candidates.csv`.

Candidate	$\log \lambda_{\mathrm{mgcv}}$	GCV	Smooth EDF	Product B
Grid interior 1	−3.3523	79.6709	7.9112	10.6287
Grid interior 2	2.1452	90.4318	3.0899	10.6108
Automatic	2.1569	90.4316	3.0814	10.6145

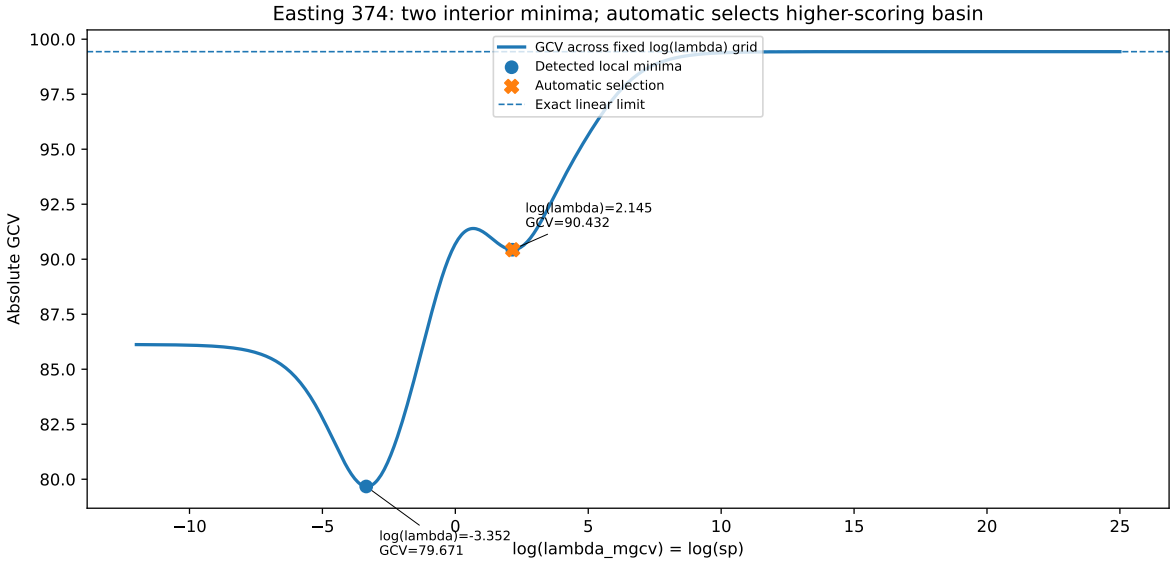


Figure 17: Two detected finite-smoothing minima for Easting 374, with the automatic parameter near the higher-scoring local basin and the exact linear-limit reference indicated. Uploaded Figure F04. Original data: `Stage3C_Full_Grid.csv`, `Stage3C_Interior_Grid_Dips.csv`, and `Stage3C_Auto_vs_Grid.csv`.

Figure 18 separates two terms in the GCV ratio. Stronger penalization reduces the smooth EDF, but raises RSS because the Northing adjustment loses flexibility. Around the more strongly penalized regime, the deterioration in fit outweighs the complexity reduction, so GCV increases toward the linear-limit score. Thus GCV is not necessarily monotone in λ, and the horizontal axis $\log(\lambda_{\text{mgcv}})$, called `log(sp)` in some original package plots, is simply the natural logarithm of the single smoothing parameter in the `mgcv` fit.

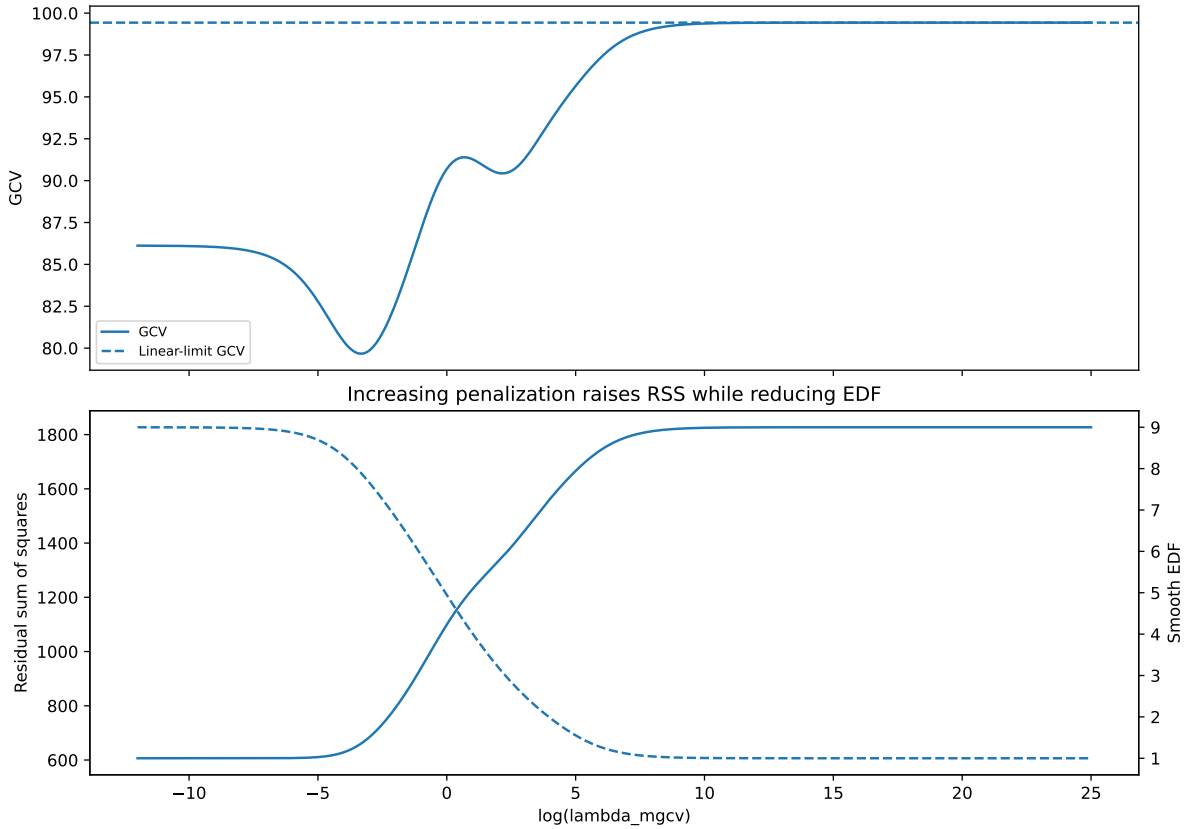


Figure 18: Easting 374: absolute GCV (upper panel), RSS and smooth effective degrees of freedom (lower panel) as functions of $\log(\lambda_{\text{mgcv}})$. The RSS and EDF directions compete in the GCV criterion. Uploaded Figure F05. Original data: `Stage3C_Full_Grid.csv` and `Stage3C_Auto_vs_Grid.csv`.

The observed difference between the two model constructions at Easting 176 further emphasizes the role of smoothing specification. The manual construction favored the linear-limit candidate (GCV $\approx$ 146.254) by approximately 0.076 relative to its best detected interior minimum, whereas the package grid slightly preferred an interior candidate (GCV $\approx$ 146.208 versus linear 146.254). The corresponding reported Product B estimates were approximately 14.688 and 13.438. This is a near-tie in the *criterion comparison* across different bases, not evidence that one coefficient is biased or that the Yield observations are inconsistent. It helps explain why a candidate-preference transition in this region occurs at a slightly different pair of adjacent Eastings under the two constructions.

Taken together, Stages 3A–3C distinguish three numerical objects: the number and arrangement of sampled criterion minima, the score achieved by an automatic optimizer relative to evaluated

alternatives, and the adjusted Product B coefficient associated with a selected smoothing regime. Their empirical relationships are heterogeneous. The completed experiments support this distinction but neither prove global GCV optima nor quantify sampling uncertainty under correlated spatial errors. A full analytic stationary-point investigation or bias assessment would require additional research beyond these results.

5.11 Descriptive multilabel anomaly taxonomy

The Stage 3C output permits a descriptive classification along three separate dimensions: sampled criterion geometry, automatic selection relative to an evaluated alternative, and the numerical difference in the unpenalized coefficient. Table 11 records the working taxonomy used in the October 20 research notes. Flags are not mutually exclusive: for example, a profile can have two local minima, an inferior automatic choice, and little change in Product B. Conversely, a single detected interior minimum does not rule out a lower-scoring linear-limit alternative. The 10^{-4} score threshold is the original numerical comparison tolerance; the coefficient-gap cutoff 0.25 and the between-minima score spread cutoff 5 are exploratory descriptive choices, *not* calibrated inferential thresholds.

Table 11: Nonexclusive descriptive anomaly taxonomy. F and G use provisional thresholds rather than calibrated error rates; category H concerns only a finite grid boundary. Source: `anomaly_taxonomy.csv`, cross-referenced to `Stage3C_Auto_vs_Grid.csv`.

Flag	Operational rule	Descriptive interpretation
`M0-M3`	0, 1, 2, or 3 interior minima	Number of detected candidate minima
`A`	At least 2 interior minima	Multiple-minimum profile
`B`	Automatic minus grid GCV > 0.0001	Inferior automatic solution
`C`	Automatic minus linear GCV > 0.0001	Inferior to linear alternative
`D`	Best interior GCV minus linear GCV > 0.0001	Linear preferred
`E`	Best interior GCV minus linear GCV < -0.0001	Interior preferred
`F`	Absolute Product B difference >= 0.25	Provisional coefficient-sensitivity flag
`G`	Minima GCV spread >= 5	Provisional widely-spaced score flag
`H`	Grid best at right edge	Strongly penalized candidate

The full set of 28 multiple-minimum Eastings is preserved in supplementary Table 16; the numerical differences across the two spline implementations at Easting 176 are presented in supplementary Figure 22. The recorded minima are sampled local turning points under finite grids, not a complete analytic enumeration of GCV stationary points. In particular, this taxonomy describes the already-completed Stage 3C results and does not implement the subsequent stationary-root analysis.

5.12 Local refinement of grid-detected GCV minima (Stage 4A)

Stage 4A refitted all 218 interior GCV minima previously detected by the 600-point `mgcv` grid. All 218 refined candidate fits met the specified local neighborhood checks; the complete validation table passed nine of nine data- and numerical-consistency checks, with no recorded fitting errors. The source-data archive includes the candidate-level outputs, full-field summary, transition table, input manifest, and validation gate. Consequently, these are *newly evaluated local refinements*

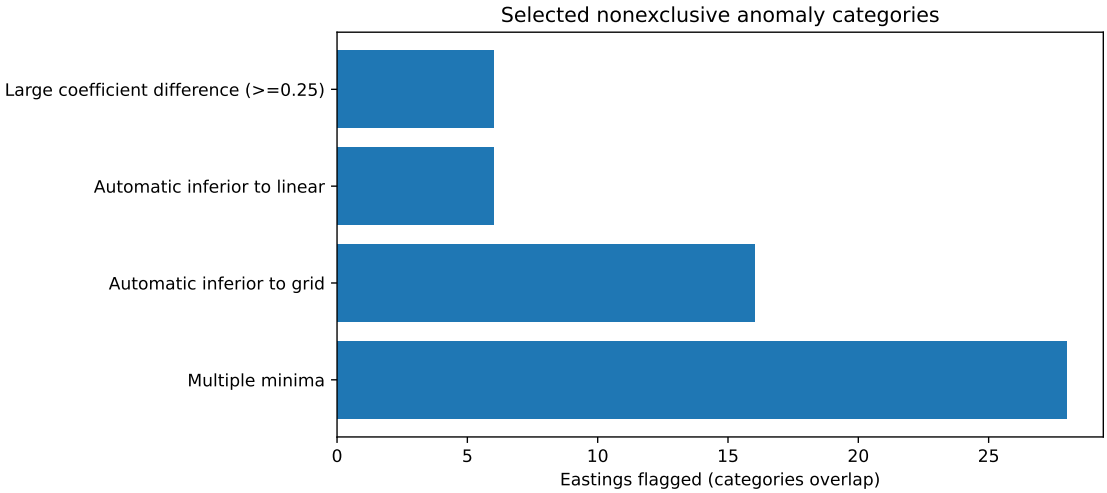


Figure 19: Counts of selected nonexclusive anomaly flags. The count of multiple-minimum Eastings is 28, automatic selections inferior to an evaluated grid alternative number 16, and six are inferior to the exact linear limit. The coefficient-difference category uses the exploratory absolute cutoff 0.25. Uploaded Figure F13. Original data: `Stage3C_Auto_vs_Grid.csv`; curated taxonomy: `anomaly_taxonomy.csv`. These are not counts of hypothesis-test rejections.

of previously identified candidates, not 218 additional independent discoveries or an exhaustive enumeration of stationary points.

The median reduction in GCV from grid-center to refined candidate was only 0.000513, with a largest reduction of 0.005259 across the 218 candidates. When the best evaluated *alternative to automatic selection* was compared Easting by Easting, its median GCV improvement was 0.000581, and the maximum was 0.004476. The corresponding median absolute change in the preferred alternative Product B estimate was 0.001544, with a maximum of 0.016189. Figure 20 displays those numerical adjustments without presenting them as changes in the unknown treatment-effect truth.

The set of 16 Eastings with inferior automatic GCV scores was *exactly the same* as under Stage 3C: 2, 28, 30, 48, 50, 52, 80, 172, 174, 214, 216, 372, 374, 376, 378, and 380. No Easting changed its preferred finite-interior-versus-linear classification: 182 favored a finite interior minimum, ten had no grid-detected interior dip, and seven favored the linear limit despite a detected interior dip. Twenty-eight Eastings continued to exhibit more than one detected interior minimum. The two close structural comparisons with absolute GCV differences under 0.10 remained at Eastings 20 and 176. Six adjacent Easting intervals changed preferred regime, as enumerated in supplementary Table 19; these are discrete changes, not demonstrated continuous crossings.

Representative score and coefficient contrasts remain qualitatively unchanged after refinement (Table 13). At Easting 216, the automatic score of approximately 171.450 exceeds the exact linear alternative 156.334 by 15.116, accompanying a Product B contrast of −0.418. At Easting 174, the score gap is 2.439, yet the Product B difference is −1.237. Conversely, the lower-GCV alternative at Easting 374 is a *refined finite interior* solution with GCV 79.670 versus automatic 90.432: the score gap is 10.762, but Product B differs by only −0.013. The revised numerical comparisons retain

Table 12: Stage 4A local-refinement verification, preserving the original Stage 3C response vectors and GAM specification. The Stage 3C and Stage 4A counts concern the same Eastings; the latter are not newly discovered cases.

Diagnostic	Verified Stage 4A result
Previously grid-detected interior minima refined	218 of 218
Candidates passing local numerical checks	218 of 218
Overall validation checks passing	9 of 9
Eastings with multiple detected interior minima	28
Eastings preferring a finite interior minimum over the exact linear limit	182
Eastings preferring the exact linear limit (no detected dip / with dip)	17 (10 / 7)
Automatic GCV inferior to an evaluated alternative by more than 10^{-4}	16 (same Eastings as Stage 3C)
Near-ties: $\lvert \mathrm{GCV}_{\mathrm{interior}} - \mathrm{GCV}_{\infty} \rvert < 0.10$	2 (20, 176)
Adjacent Easting intervals changing preferred regime	6
Median grid-minus-refined GCV per detected candidate	0.000513
Largest grid-minus-refined GCV per detected candidate	0.005259
Median best-alternative GCV improvement per Easting	0.000581
Largest best-alternative GCV improvement per Easting	0.004476
Median $\lvert \widehat{\beta}_{B,\mathrm{alt,4A}} - \widehat{\beta}_{B,\mathrm{alt,3C}} \rvert$	0.001544
Largest corresponding absolute Product B change	0.016189

Table 13: Representative automatic fits and best evaluated alternatives after Stage 4A refinement. D_{4A} is automatic minus alternative GCV, while $\Delta_{B,4A}$ is the corresponding signed Product B coefficient difference. A higher GCV score does not imply a larger coefficient difference.

Easting	$\mathrm{GCV}_{\mathrm{auto}}$	$\mathrm{GCV}_{\mathrm{alt}}$	D_{4A}	$\Delta_{B,4A}$	Alternative
174	145.471	143.032	2.439	-1.237	Linear
214	158.473	153.498	4.975	-0.358	Linear
216	171.450	156.334	15.116	-0.418	Linear
374	90.432	79.670	10.762	-0.013	Refined interior
376	92.319	82.268	10.051	+0.290	Refined interior

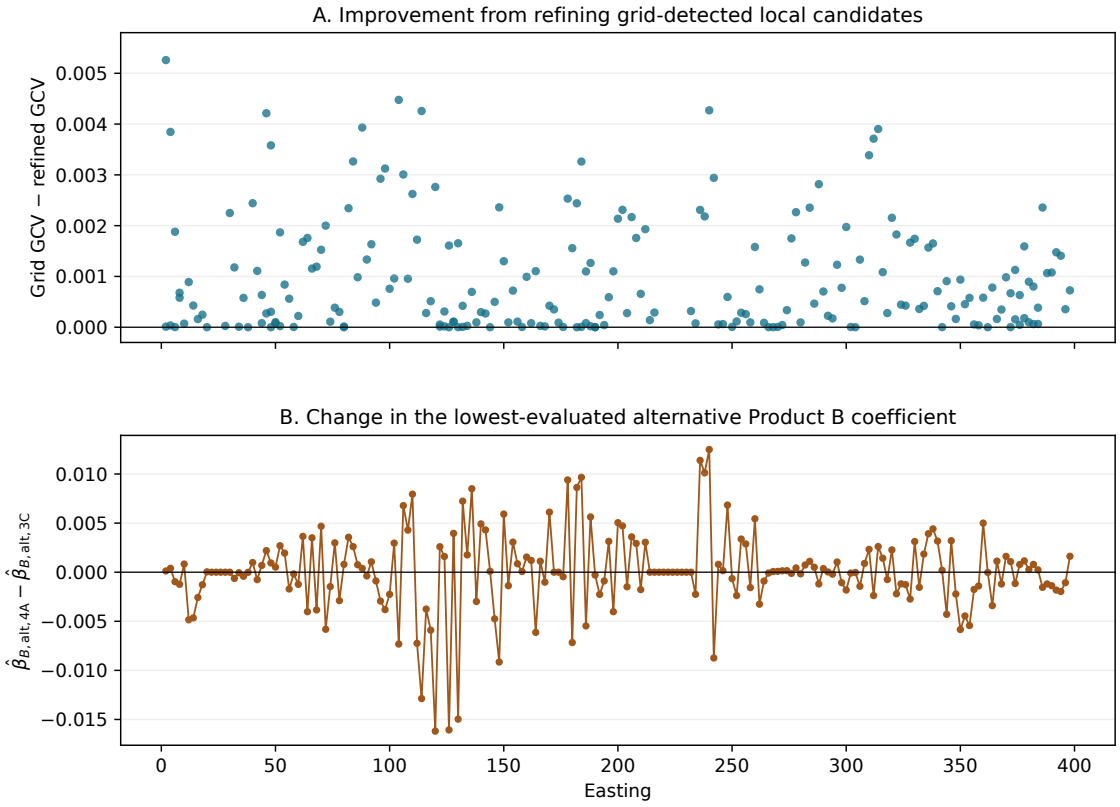


Figure 20: Stage 4A refinement effects. Top: GCV decrease obtained by minimizing within each previously detected grid bracket (all 218 candidates). Bottom: change in the Product B coefficient of the best evaluated alternative after refinement versus the Stage 3C benchmark (199 Eastings). The former compares a grid candidate and a local optimizer within one basin; the latter can be zero when the exact linear limit was already preferred. Computed from `Stage4A_Refined_Interior_Minima.csv` and `Stage4A_vs_Stage3C_Comparison_Audit.csv`.

six inferior automatic selections with absolute Product B discrepancies of at least 0.25, including three at least 0.50. Thus, validation of the local candidate locations strengthens the conclusion that GCV optimization discrepancies and scientific-coefficient discrepancies need not have proportional magnitudes.

The reference definition matters for the field-wide summary. From Stage 3C, the median absolute difference between automatic and lowest-*grid* product coefficients is 0.002198. The Stage 4A median absolute automatic-versus-*best-alternative* difference (excluding the automatic fit from the candidates) is approximately 7.9×10^{-9}, and the automatic-versus-*best-evaluated* difference (including it) is 5.1×10^{-9}. The corresponding automatic-versus-*exact-linear* median absolute difference is 1.184991. Most small grid discrepancies therefore do not establish an optimization failure: at those locations the automatic solution and the continuously refined preferred candidate are numerically almost identical. The 16 flagged locations remain genuine counterexamples to an assertion that automatic GCV is always lowest among the evaluated alternatives. None of these numerical contrasts estimates sampling uncertainty or treatment-effect bias.

The refinement results establish stability of the reported numerical comparisons to *within-bracket refinement of the detected minima*. They do not establish global GCV optimality, explain the mechanism of each automatic optimizer stop, provide sampling uncertainty for the coefficient differences, or infer a true local agricultural feature from the coefficient trajectory.

Deferred sensitivity analysis. *Revision placeholder:* A separate set of first-stage response-processing sensitivity analyses will be evaluated after the core Stage 1–4A manuscript is finalized. These results have deliberately not been incorporated into the numerical claims, figures, tables, or conclusions of this version.

6 Discussion, Limitations, and Implications

Draft placeholder. This section is included in the proposed table of contents but has not yet been written. No new analyses are asserted here.

7 Conclusion

Draft placeholder. This section is included in the proposed table of contents but has not yet been written. No new analyses are asserted here.

A Supplementary Numerical Exhibits

These supplementary displays preserve the remaining uploaded figures and curated CSV tables while avoiding repetition of the principal results in Section 5. Values in the curated tables were exported as rounded presentation summaries; higher precision is retained in the canonical Stage 1–3C and Stage 4A CSVs in `source_data/`. Stage 4A provides verified fixed-parameter refits and local optimization within detected brackets, but no analytic stationary-point enumeration or simulation-based inference.

A.1 Stage workflow summary and early comparisons

Table 14: Compact chronological index to the completed experiments. This table is retained only for reproducibility; it does not represent a further statistical result. Curated source: `stage_summary.csv`; underlying analyses: `Stage1B_Summary.csv`, `Manual_Full_Field_Summary.csv`, `Stage3A_Auto_vs_Grid.csv`, `Stage3B_Linear_Screen_and_Trajectories.csv`, and `Stage3C_Auto_vs_Grid.csv`.

Stage	Experiment	Completed result	Question answered
1A	Full-data audit	4,776 records; 199 Eastings; 24 passes	What is the design?
1B	Slice reconciliation	72 matches	Are original slices unchanged?
2	Manual GCV	79,600 grid scores; 27 multi-minimum Eastings	What is the manually built GCV landscape?
3A	mgcv pilot	3,600 fixed fits	Can automatic selection miss better grid candidates?
3B	Automatic full field	199 automatic fits; 6 worse than linear	Where does a known lower GCV alternative exist?
3C	mgcv full grid	119,400 fits; 16 inferior selections; 28 multi-minimum Eastings	What is the full evaluated optimizer gap?

Table 15: The original three Eastings compared across manual and package-based candidate definitions. The exact linear-limit score is shared, while the smoothing parameter scales and finite-spline fits differ. Rounded source: `original_slice_manual_and_mgcv.csv`; original files: `Manual_Full_Field_Summary.csv` and `Stage3C_Auto_vs_Grid.csv`.

Easting	Manual interior	Exact linear	Automatic `mgcv`	Best package grid
214	158.422	153.498	158.473	153.498
216	171.417	156.334	171.450	156.334
218	none detected	160.229	160.229	160.229

The full six-Easting pilot comparison is already Table 6; its original condensed CSV is preserved in `source_data/curated_tables/six_easting_pilot_comparison.csv`. Table 8 and Table 9 likewise incorporate the complete original 16-Easting and selected two-Easting curated comparisons, respectively; their rounded CSVs are retained with the project.

A.2 Minimum counts and multiple-minimum locations



Figure 21: Distribution of zero, one, two, or three grid-detected interior minima under the manual spline and `mgcv` constructions. This supplemental histogram visualizes Table 7 without imposing a common smoothing-parameter scale. Uploaded Figure F09. Original data: `Manual_Full_Field_Summary.csv` and `Stage3C_Auto_vs_Grid.csv`; curated table: `counts_of_interior_minima.csv`.

Table 16: All 28 Eastings with at least two grid-detected interior minima under the package-based model. Flags refer only to the finite numerical grid; negative small optimizer gaps are possible because the automatic fit is not confined to sampled grid points. Curated source: `twenty_eight_multiple_minimum_eastings.csv`; original sources: `Stage3C_Auto_vs_Grid.csv` and `Stage3C_Interior_Grid_Dips.csv`.

Easting	Interior minima	Auto worse than grid?	GCV gap
2	2	Yes	6.581
4	2	No	-0.004
6	2	No	-0.002
8	2	No	-0.001
44	2	No	-0.001
46	2	No	-0.004
48	3	Yes	1.675
50	2	Yes	0.961
52	2	Yes	1.244
80	2	Yes	0.017
122	2	No	-0.000
124	2	No	-0.000
126	2	No	-0.002
128	2	No	-0.000
130	2	No	-0.002
132	2	No	-0.000

Table 16 continued

Easting	Interior minima	Auto worse than grid?	GCV gap
182	2	No	-0.002
184	2	No	-0.003
186	2	No	-0.001
188	2	No	-0.001
190	2	No	-0.000
372	2	Yes	0.587
374	2	Yes	10.761
376	2	Yes	10.050
378	2	Yes	8.169
380	2	Yes	9.132
382	2	No	-0.001
384	2	No	-0.000

A.3 Detailed localized comparisons

Table 17: Easting 176: the manual and package finite-smoothing criterion profiles have different slight preferences relative to a common exact linear-limit GCV. Preferred Product B estimates differ because the basis and penalty constructions differ. Curated source: `easting_176_comparison.csv`; original sources: `Manual_Full_Field_Summary.csv` and `Stage3C_Auto_vs_Grid.csv`.

Model	Interior GCV	Linear-limit GCV	Preferred Product B
Manual	146.3298	146.2541	14.6881
mgcv	146.2077	146.2541	13.4384

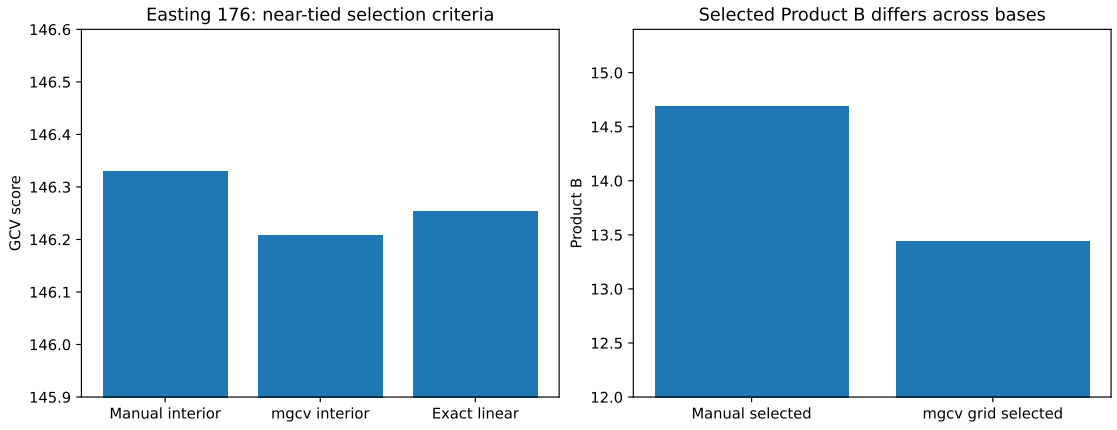


Figure 22: Near-tied GCV preferences at Easting 176 under different spline constructions, with differing preferred adjusted Product B estimates. This is basis-dependent criterion sensitivity, not a proof of bias. Uploaded Figure F12. Original data: `Manual_Full_Field_Summary.csv` and `Stage3C_Auto_vs_Grid.csv`; curated table: `easting_176_comparison.csv`.

The more detailed candidate-level comparison for Easting 374 appears in Table 10, and the

Table 18: Selected points along the Easting 374 GCV profile, showing the lower-scoring flexible minimum and the rise toward a high-penalty plateau. These are sampled points, not a proof of global minima. Curated source: `easting_374_criterion_along_grid.csv`; original sources: `Stage3C_Full_Grid.csv` and `Stage3C_Interior_Grid_Dips.csv`.

Grid position	$\log \lambda_{\text{mgcv}}$	GCV	Smooth EDF
Grid left edge	-12.000	86.120	9.000
First interior dip	-3.352	79.671	7.911
Second interior dip	2.145	90.432	3.090
Grid right edge	25.000	99.434	1.000

Table 19: Adjacent Easting intervals in which the Stage 4A criterion-preferred regime changes. These are observed discrete transitions, not claims about a continuous crossing or bifurcation.

From	To	Regime (from)	Regime (to)	$\Delta\widehat{\beta}_{B,\text{auto}}$	$\Delta\widehat{\beta}_{B,\text{best}}$
20	22	Interior	Linear	+0.234	+0.234
30	32	Linear	Interior	+0.096	-0.108
170	172	Interior	Linear	+0.349	+1.043
174	176	Linear	Interior	+0.151	-1.086
212	214	Interior	Linear	-0.357	+0.002
232	234	Linear	Interior	-1.491	-1.491

GCV/RSS/EDF visualization appears in Figure 18. The curated source files and their original inputs are both included to permit manual checks of the reported values.

A.4 Stage 4A transition diagnostics and run provenance

The supplementary file `figures/stage4A_original_diagnostics.pdf` preserves the original five-page R output, including the per-Easting criterion gap, coefficient differences, structural gaps, trajectories, and selected GCV profiles. The Stage 4A full-field CSV and candidate CSV retain unrounded values; the validation gate records all nine passing checks.

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